
Chapter 9

Operational Amplifiers

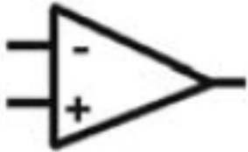
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Outline

- **General Consideration**
- Single-Stage OTA
- Two-Stage OTA
- Gain Boosting Techniques
- Common-Mode Feedback
- Input Range Limitations
- Slew Rate
- Power Supply Rejection

What is an operational amplifier?

Opamp	OTA
Operational amplifier	Operational Transconduct. amplifier
	
$A_v = \frac{V_{OUT}}{V_{IN}}$	$A_g = \frac{i_{OUT}}{V_{IN}}$
$A_v =$	$= A_g R_L$
$R_{in} = \infty$	$R_{in} = \infty$
$R_{out} = 0$	$R_{out} = \infty$

- Operational amplifier = Opamp
- Difficult to obtain a low output resistance in CMOS
- OTAs are more often used in CMOS process

Applications of OPA/OTA

- An integral part of many analog/mixed-signal system
 - Can be provided as a single chip
 - OP07, OP27, OP37, LM341.....
 - Today , OPA/OTA are more often used as IPs in large scale ICs
- OPA/OTA are applied everywhere
 - ADC/DAC
 - Filter
 - DC/DC

Specification parameters

- Gain
- Small-Signal Bandwidth
- Large-Signal Bandwidth
- Output Swing
- Linearity
- Noise (IRN)
- Offset
- Supply Rejection

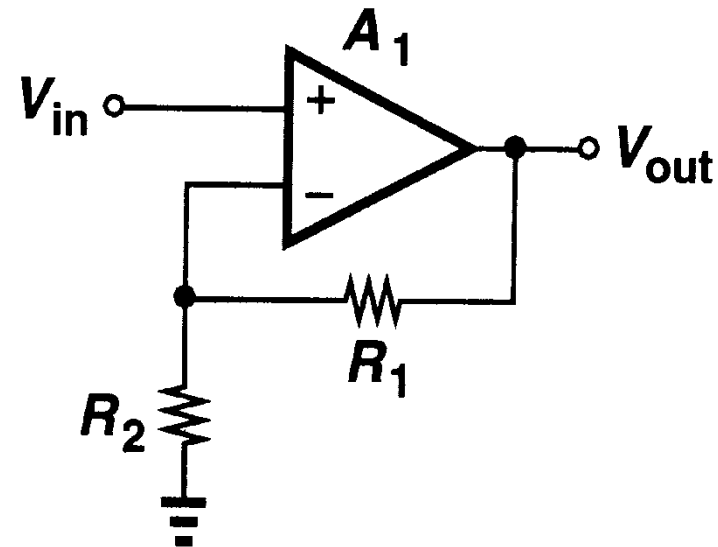


Example 9.1: Determine the **gain** of op amps

The circuit is designed for a nominal gain of 10, i.e., $1+R_1/R_2=10$. Determine the minimum value of A_1 for a gain error of 1% ($1/\beta=10$).

$$\frac{V_{out}}{V_{in}} = \frac{1}{\beta} \frac{A_1 \beta}{1 + \beta A_1} = \frac{A_1}{1 + \frac{R_2}{R_1 + R_2} A_1}$$

$$\frac{V_{out}}{V_{in}} \approx \left(\frac{1}{\beta} \right) \left(1 - \frac{1}{\beta A_1} \right)$$



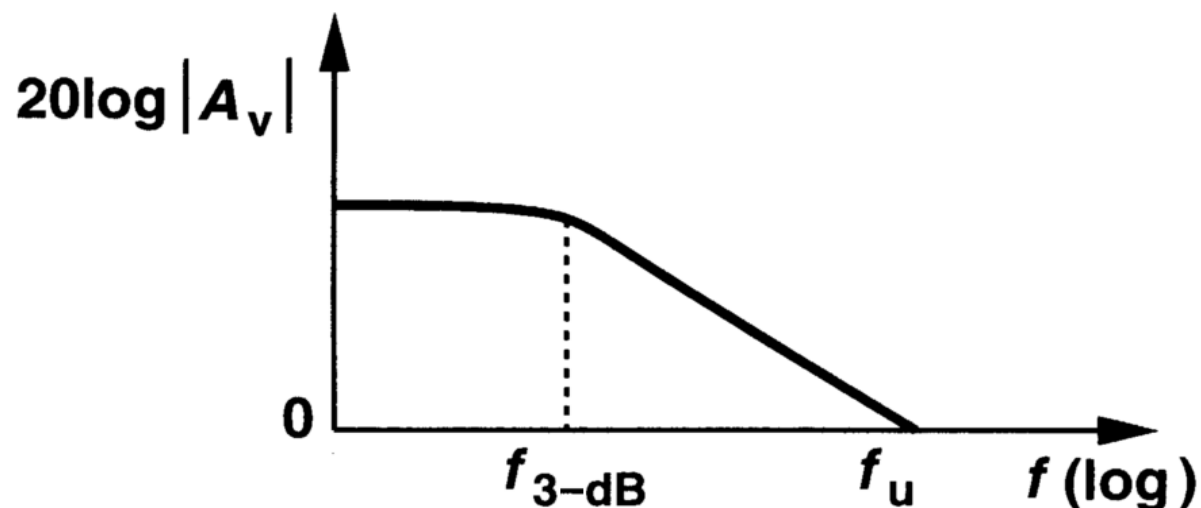
To achieve a gain error less than 1% ,we must have $A_1 > 1000$

The gain of open-loop op amp depends on the required precision in its closed-loop application!



Small-signal bandwidth

- Under the condition of steady-state sine input
 - Small signal gain decreases at high frequency



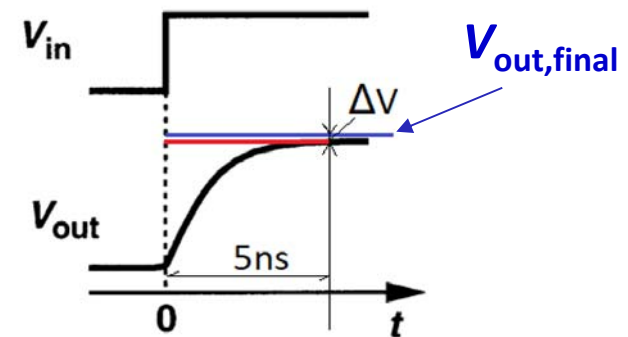
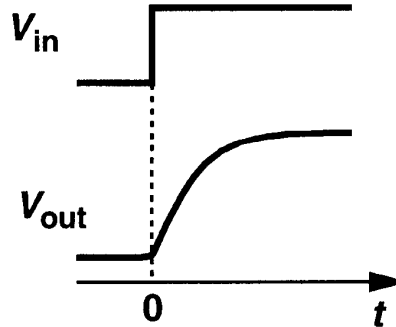
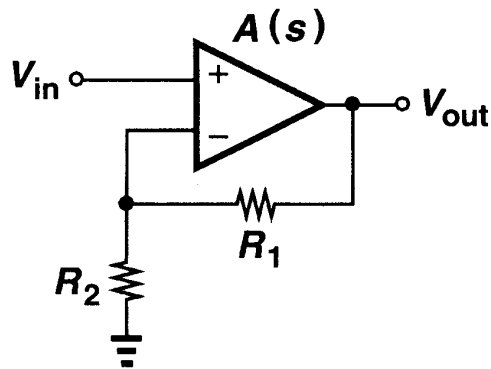
f_u : Unity-gain frequency

f_{3-dB} : -3dB frequency

The speed is determined by GBW

Example 9.2:

For the non-inverting feedback structure ($1+R_1/R_2 \approx 10$) with a single-pole opamp, if we want to achieve a settling error of less than 1% within 5ns, please determine the minimum gain-bandwidth product (GBW) of the Amplifier.

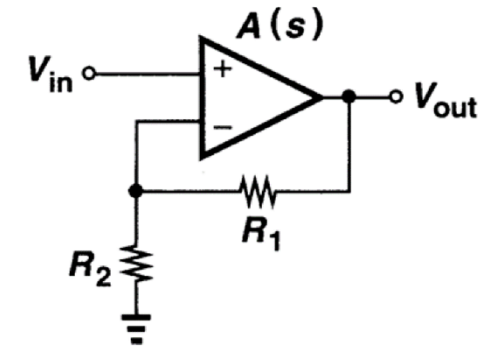


$$\frac{\Delta V}{V_{out,final}} \leq 1\%$$



$$A(s) = \frac{A_0}{(1 + s / \omega_0)}, \quad \beta = \frac{R_2}{R_2 + R_1}, \quad \frac{V_{out}}{V_{in}}(s) = \frac{1}{\beta} \frac{\beta A(s)}{1 + \beta A(s)}$$

$$\text{GBW} = A_0 \omega_0$$



Low-Frequency Gain

$$\frac{V_o}{V_i}(s) = \frac{1}{\beta} \frac{\beta A_0}{1 + \beta A_0} \cdot \frac{1}{1 + s(1 + \beta A_0)\omega_0}, \quad \tau = \frac{1}{(1 + \beta A_0)\omega_0} \approx \frac{1}{\beta A_0 \omega_0}$$

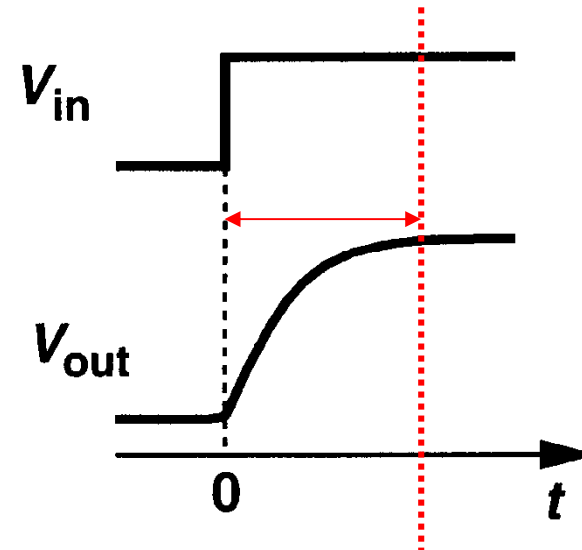
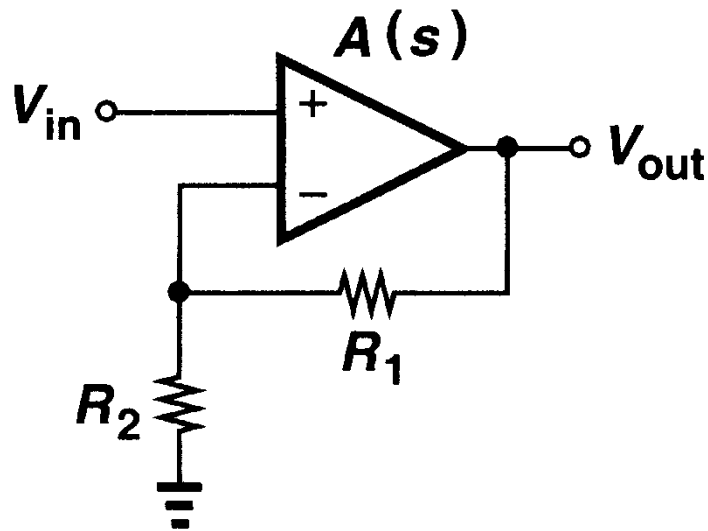
Final Value

For $V_{in} = V_d u(t) \Rightarrow V_o(t) = V_d \frac{1}{\beta} \frac{\beta A_0}{1 + \beta A_0} \left(1 - e^{-\frac{t}{\tau}} \right)$

$$e^{-\frac{t}{\tau}} = e^{-t\beta A_0 \omega_0} < 1\% \Rightarrow \text{GBW} = A_0 \omega_0 = 9.21 \text{ Grad} / s \approx 1.47 \text{ GHz}$$

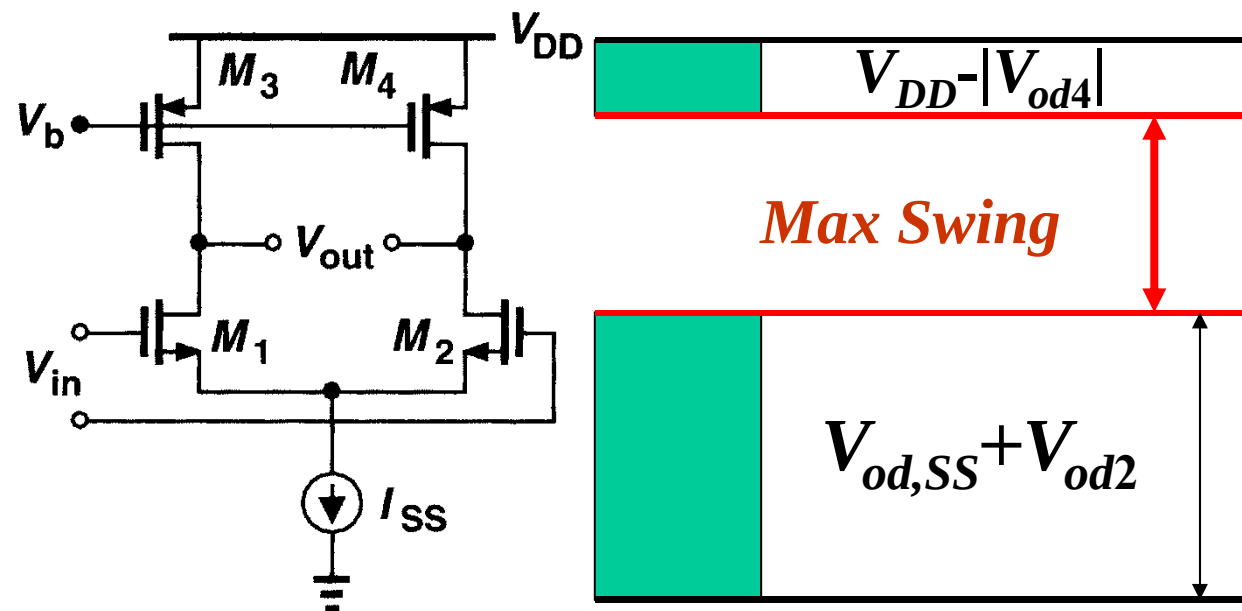
Large-signal bandwidth

- Under the condition of large step input signal
 - How fast does the output settling to final value
 - The settling process is nonlinear
 - Important especially in sampling systems



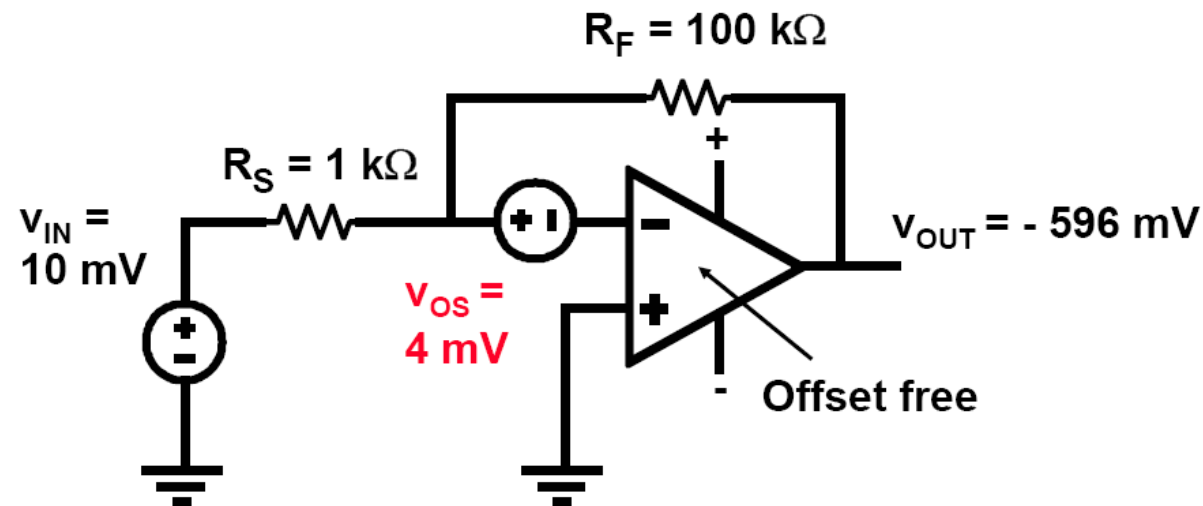
Output swing

- An important spec. related to dynamic range
 - The maximum/minimum output voltage at where all MOS operate in saturation.
 - Often determined by the overdrive voltage of MOS



Offset

- When the differential input voltage is 0
 - Output is not at its ideal DC operation point, the voltage difference is defined as output offset
 - Often referred to input → **input offset voltage**
 - $V_{os,in} = V_{os,out}/A_{v0}$



The gain is 59 instead of 100 !



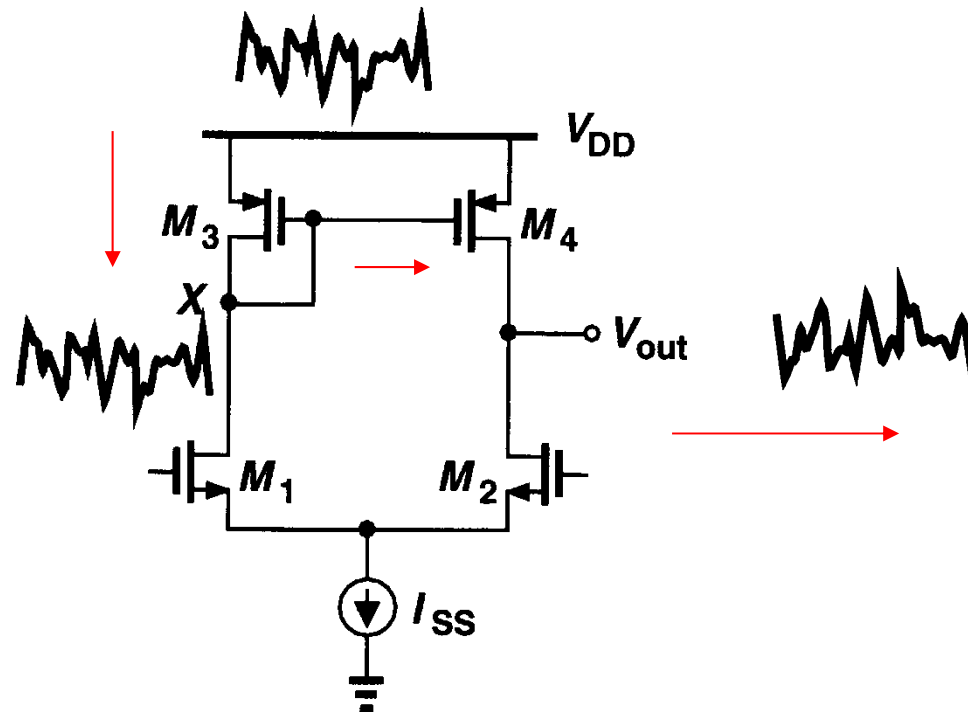
Linearity

- The gain of real amplifiers varies significantly with the input amplitude
- Non linearity can be rejected with feedback.



Supply rejection

- The noise carried in power supply is coupled to the output
 - Fully differential topologies are preferred



Outline

- General consideration
- **Single-stage OTA**
 - **Differential pair**
 - Telescope OTA
 - Folded cascode OTA
- Two-Stage OTA
- Gain boosting technique
- Common-mode feedback
- Input range limitations
- Slew Rate
- Power supply rejection

Differential pair with active current mirror(1)

- Large signal performance

- Power Diss: $I_{D5} * V_{DD}$

- Input CM range

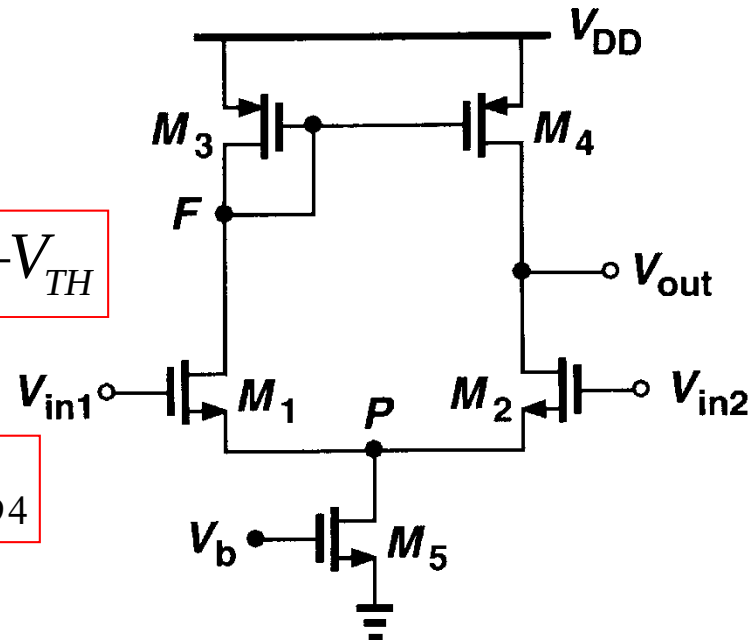
$$V_{OD5} + V_{GS1} \leq V_{in,CM} \leq V_{DD} - |V_{GS3}| + V_{TH}$$

- Output Swing

$$V_{OD5} + V_{OD2} \leq V_{out} \leq V_{DD} - |V_{OD4}|$$

- Output DC point

$$V_{out,DC} = V_{DD} - |V_{GS3}|$$



Differential pair with active current mirror(2)

- **Small-signal performance**

- **Low-frequency gain**

$$A_{v0} = g_{m1,2} (r_{o2} \parallel r_{o4})$$

- **Open-loop transfer function**

$$A_v(s) = A_{v0} \cdot \frac{1 + s/(2\omega_{p2})}{(1 + s/\omega_{p1})(1 + s/\omega_{p2})}$$

$$\omega_{p1} \approx \frac{1}{(r_{oN} \parallel r_{oP})C_L}$$

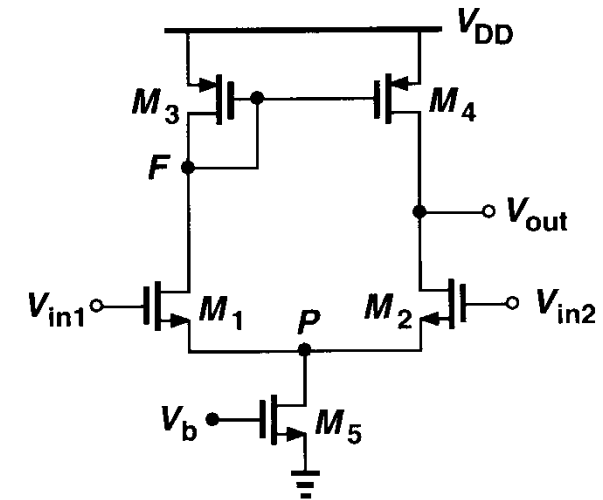
$$\omega_{p2} \approx \frac{g_{mP}}{C_F}$$

- **3-dB bandwidth**

$$BW \approx \frac{1}{2\pi \cdot (r_{oN} \parallel r_{oP})C_L} \quad (\text{Hz})$$

- **Gain bandwidth product**

$$GBW \approx A_{v0} \cdot BW = \frac{g_{m1,2}}{2\pi \cdot C_L} \quad (\text{Hz})$$



Fully differential pair with current source load (1)

- Large signal performance

- Power Diss: $I_{SS} * V_{DD}$

- Input CM range

$$V_{OD5} + V_{GS1} \leq V_{in,CM} \leq V_{out,DC} + V_{TH}$$

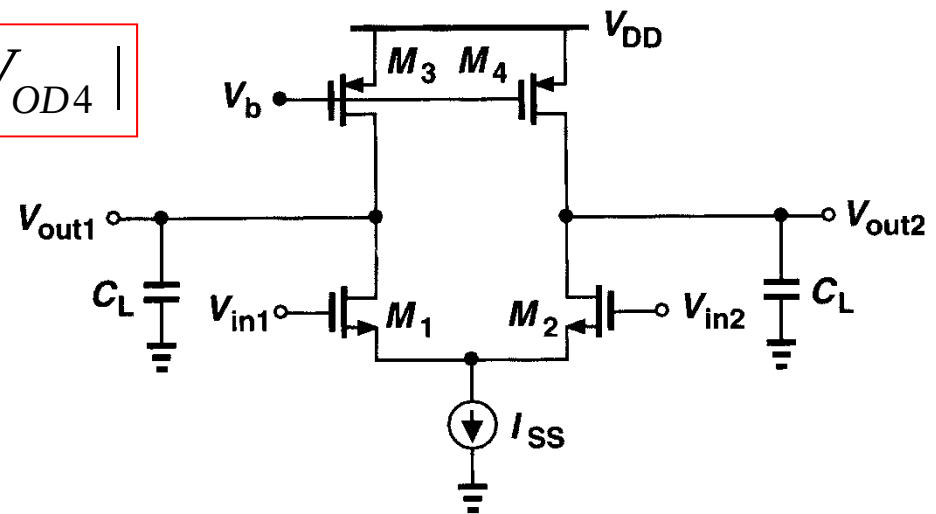
- Output Swing

$$V_{OD,SS} + V_{OD2} \leq V_{out2} \leq V_{DD} - |V_{OD4}|$$

- Output DC point

- Not sure

- Need CMFB



Fully Differential pair with current source load (2)

- **Small-signal performance**

- **Low-frequency gain**

$$A_{v0} = g_{m1,2} (r_{o2} \parallel r_{o4})$$

- **Open-loop transfer function**

$$A_v(s) = A_{v0} \cdot \frac{1}{(1 + s / \omega_{p1})}$$

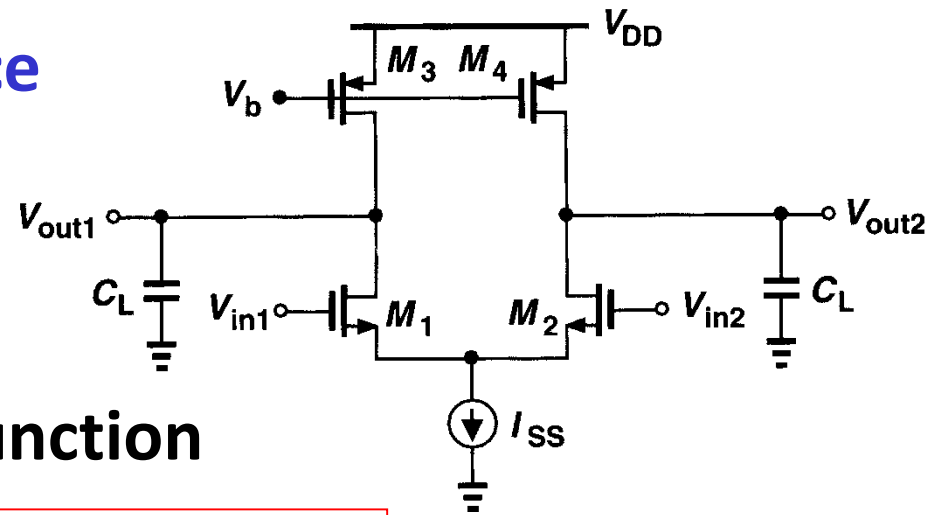
$$\omega_{p1} \approx \frac{1}{(r_{oN} \parallel r_{oP})C_L}$$

- **3-dB bandwidth**

$$BW \approx \frac{1}{2\pi \cdot (r_{oN} \parallel r_{oP})C_L} \quad (\text{Hz})$$

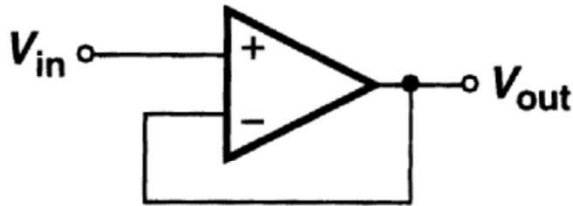
- **Gain bandwidth product**

$$GBW \approx A_{v0} \cdot BW = \frac{g_{m1,2}}{2\pi \cdot C_L} \quad (\text{Hz})$$



Ex. 9.3 Unity Gain Configuration

- Calculate input common range and closed-loop output impedance

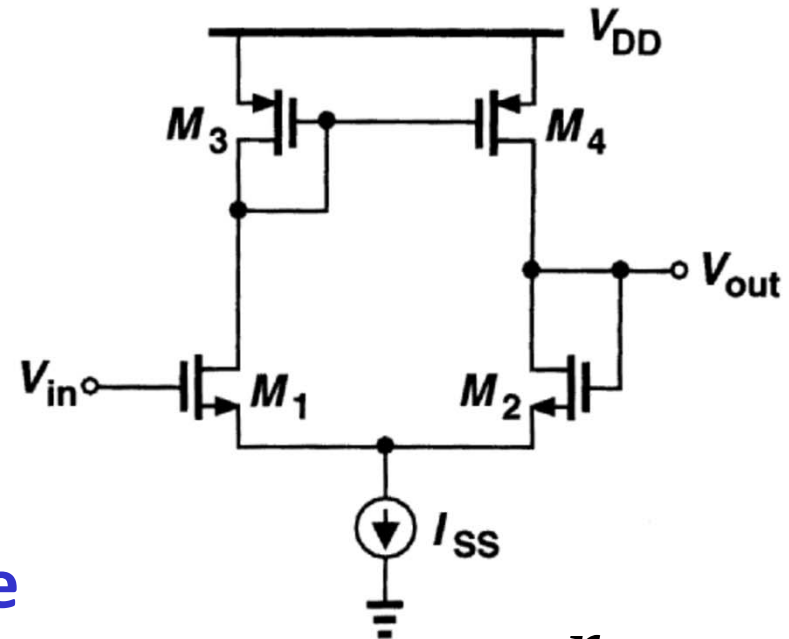


$$V_{OD,SS} + V_{GS1} \leq V_{in,CM} \leq V_{DD} - |V_{GS3}| + V_{TH}$$

- Open-loop output resistance

$$r_{out,open} = r_{o2} \parallel r_{o4}$$

$$\beta = 1 \quad A_{v0} \cdot \beta = g_{m1,2} (r_{o2} \parallel r_{o4})$$



$$r_{out,cl} = \frac{r_{out,open}}{1 + \beta A_{v0}} \approx 1 / g_{m2}$$



Design procedure for differential pair

- Design procedure

- (1) Power diss $\rightarrow I_D$

- (2) Output swing or output DC point $\rightarrow V_{OD}$

- (More practically, I_D and V_{OD} should be co-designed according to the required GBW and noise specifications)*

- (3) V_{OD} and $I_D \rightarrow W/L$

- (4) V_{OD} and $I_D \rightarrow$ bias voltage or current

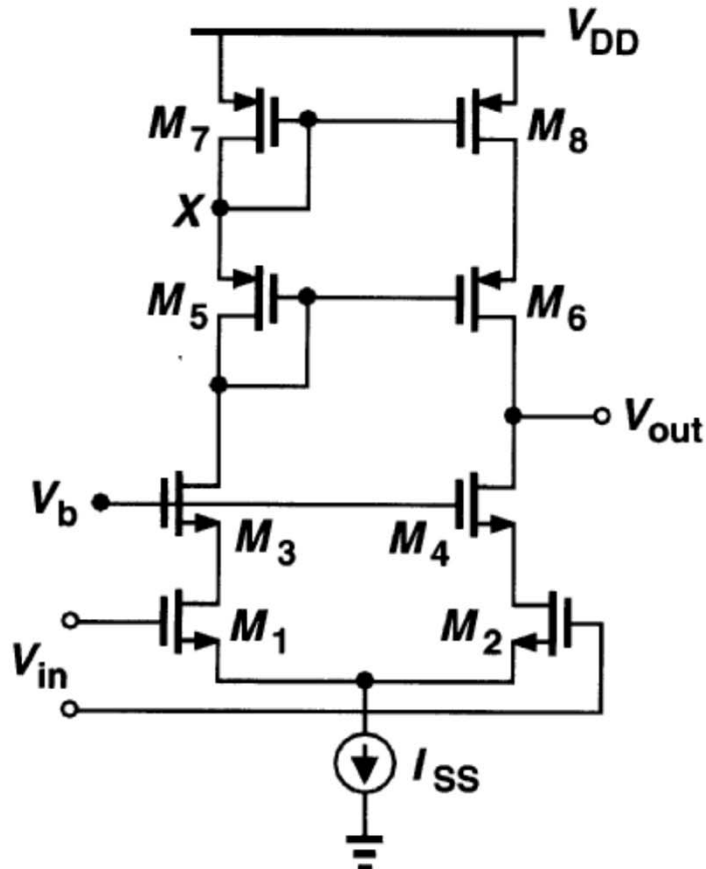
- (5) **Verify** the gain

- (6) Design iteration: **adjusting L while keeping W/L constant**

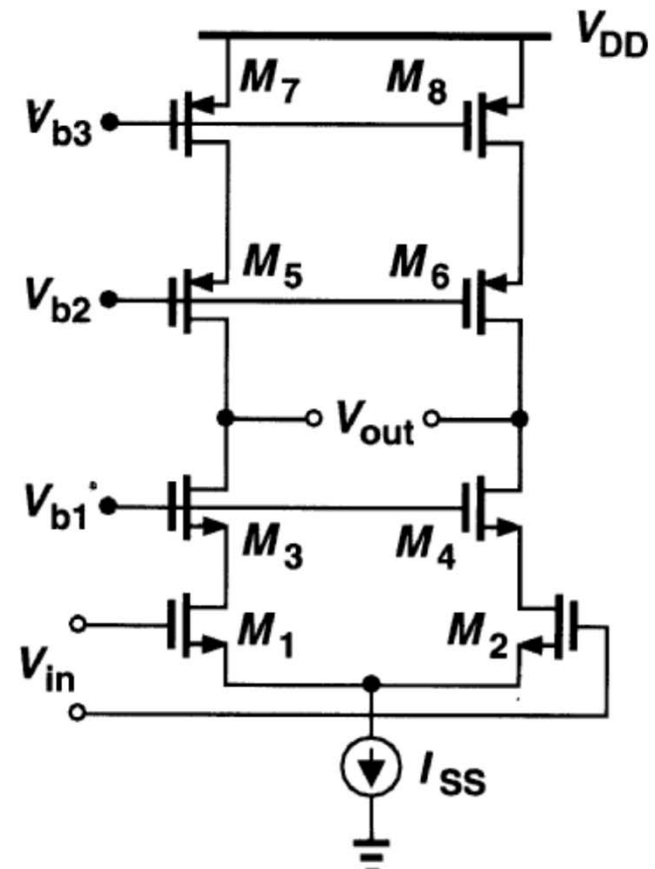
Outline

- General consideration
- **Single-stage OTA**
 - Differential pair
 - **Cascode (Telescope) OTA**
 - Folded cascode OTA
- Two-Stage OTA
- Gain boosting technique
- Common-mode feedback
- Input range limitations
- Slew Rate
- Power supply rejection

Cascode OTAs



Single-ended output



Fully differential



Single-ended output cascode OTA

- Large signal performance

- Power diss.: $I_{SS} * V_{DD}$

- Input CM range

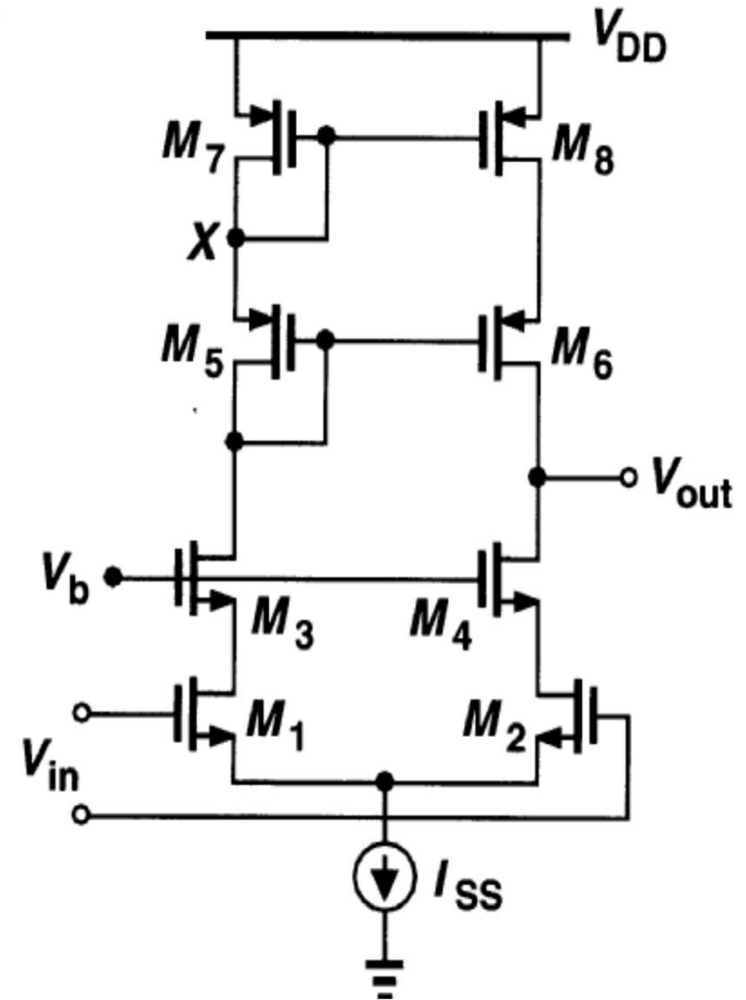
$$V_{OD,SS} + V_{GS1} \leq V_{in,CM} \leq V_b - V_{GS3} + V_{TH}$$

- Output DC point

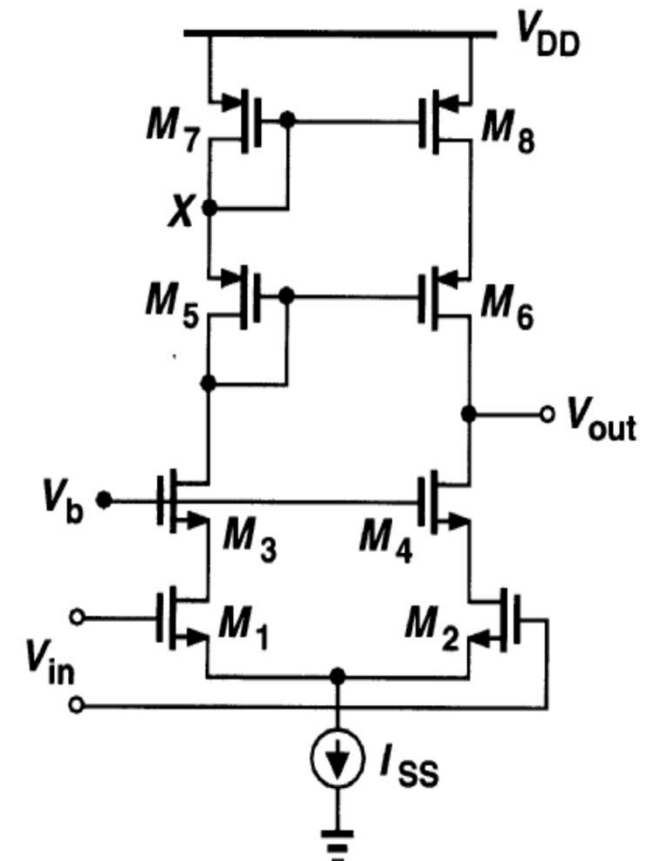
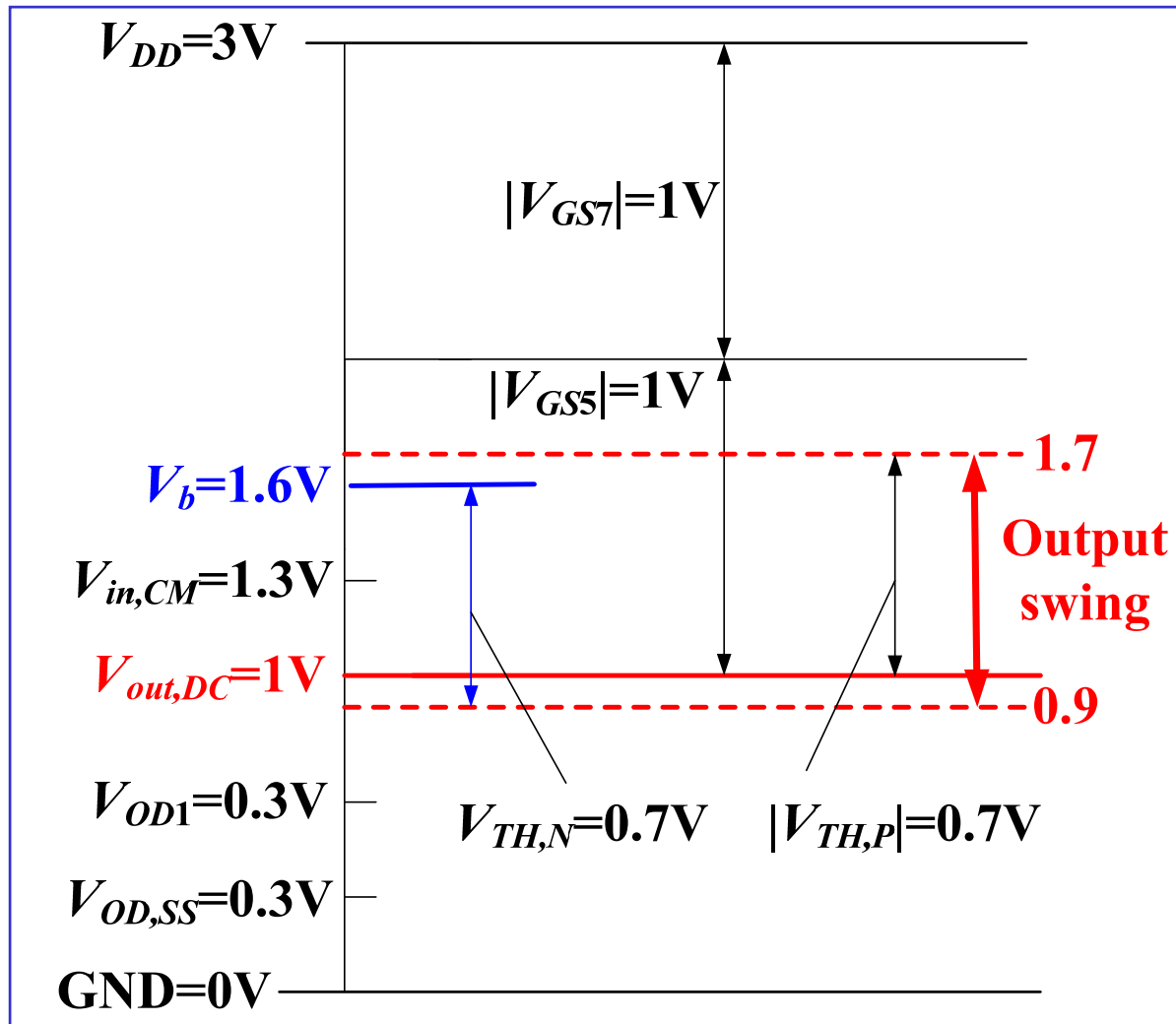
$$V_{out,DC} = V_{DD} - |V_{GS5}| - |V_{GS7}|$$

- Output Swing

$$V_{OD,SS} + V_{OD2} + V_{OD4} \leq V_{out} \leq V_{DD} - |V_{GS7}| - |V_{GS5}| + V_{TH}$$



Example of output swing for Cascode OTA



Large-swing Single-ended output cascode OTA

- Large signal performance

- Power Diss: $I_{SS} * V_{DD}$

- Input CM range

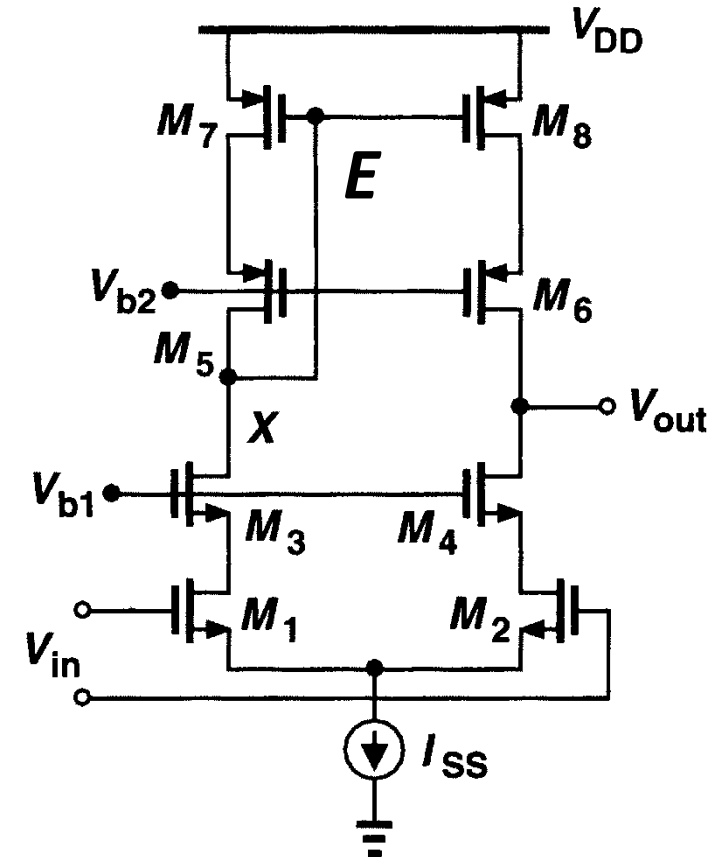
$$V_{OD,SS} + V_{GS1} \leq V_{in,CM} \leq V_{b1} - V_{GS3} + V_{TH}$$

- Output DC point

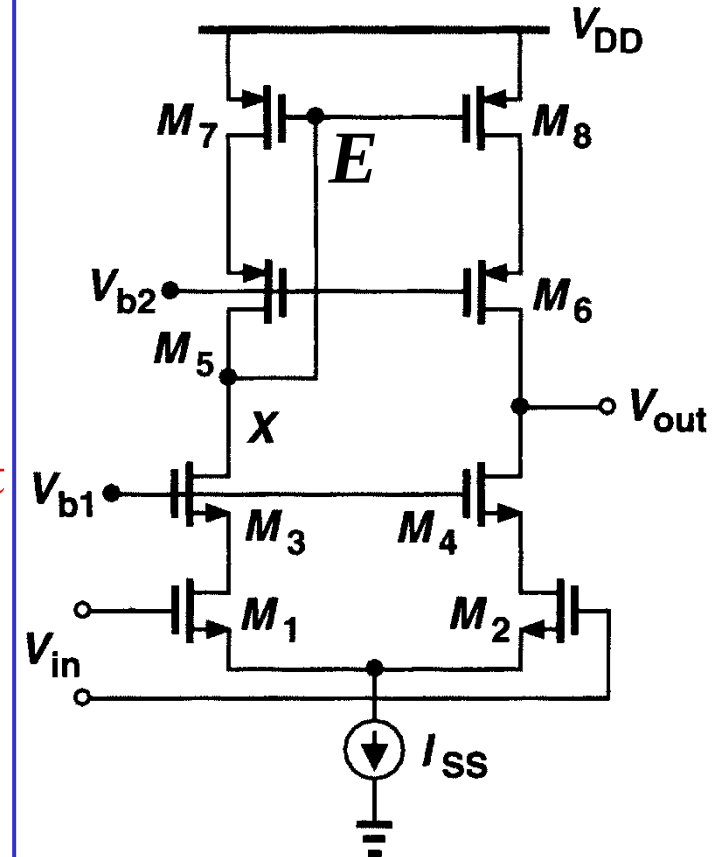
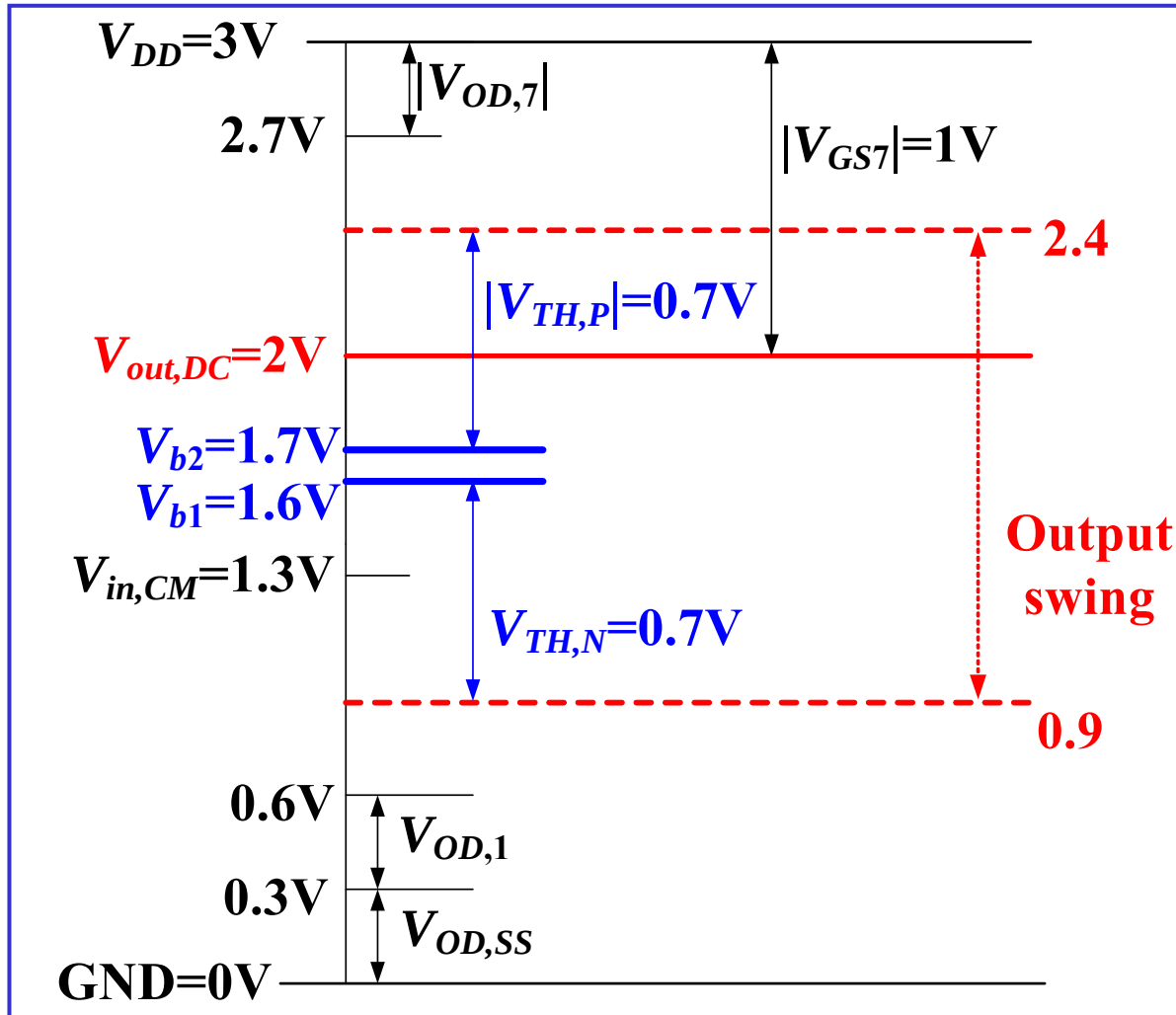
$$V_{out,DC} = V_{DD} - |V_{GS7}|$$

- Output Swing

$$V_{b1} - V_{TH4} \leq V_{out} \leq V_{b2} + |V_{TH6}|$$



Example of output swing for Cascode OTA



Cascode OTA Small-signal performance

– Low-frequency gain

$$A_{v0} = g_{mN} \cdot [(g_{mN} \cdot r_{oN}^2) \parallel (g_{mP} \cdot r_{oP}^2)]$$

– Open-loop transfer function

$$A_v(s) = A_{v0} \cdot \frac{(1 + s / 2\omega_{p2})}{(1 + s / \omega_{p1})(1 + s / \omega_{p2})}$$

$$\omega_{p2} \approx \frac{g_{mP}}{C_E}$$

– 3-dB bandwidth

$$\omega_{p1} \approx \frac{1}{[(g_{mN} \cdot r_{oN}^2) \parallel (g_{mP} \cdot r_{oP}^2)]C_L}$$

-Gain bandwidth product

$$BW \approx \frac{\omega_{p1}}{2\pi} \quad (\text{Hz})$$

$$GBW \approx A_{v0} \cdot BW = \frac{g_{m1,2}}{2\pi \cdot C_L} \quad (\text{Hz})$$



Fully differential Cascode OTA (1)

- Large signal performance

- Power Diss: $I_{SS} * V_{DD}$

- Input CM range

$$V_{OD,SS} + V_{GS1} \leq V_{in,CM} \leq V_{b1} - V_{GS3} + V_{TH}$$

- Output DC point

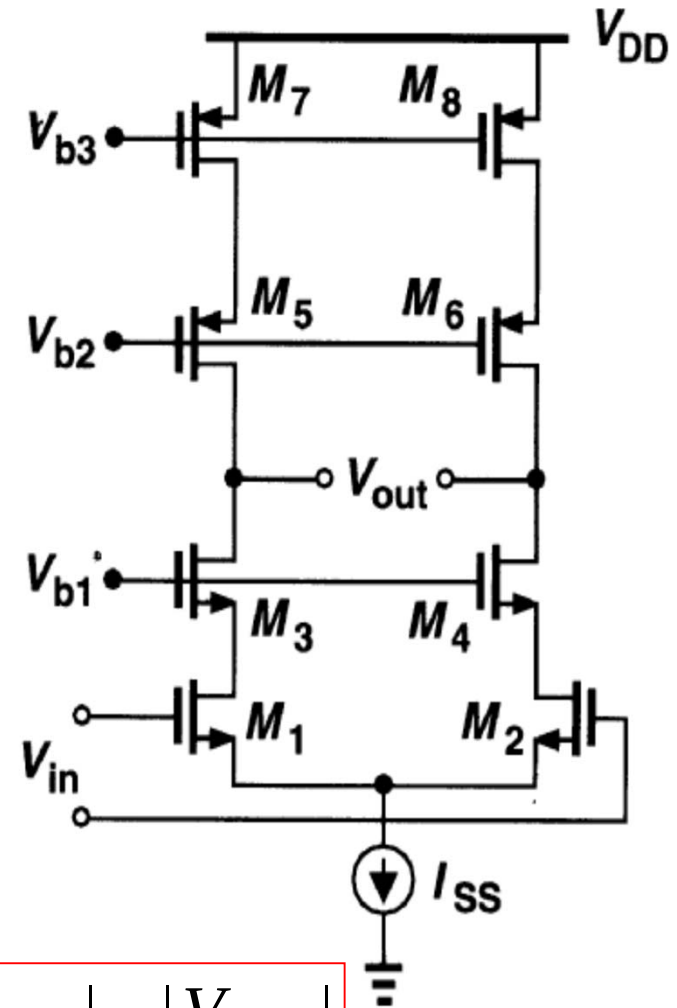
- Not sure

- Need CMFB

- Output Swing

$$V_{b1} - |V_{TH}| \leq V_{out1,2} \leq V_{b2} + |V_{THP}|$$

$$V_{OD,SS} + V_{OD1} + V_{OD3} \leq V_{out1,2} \leq V_{DD} - |V_{OD6}| - |V_{OD8}|$$



Fully differential Cascode OTA (2)

- **Small-signal performance**

- **Low-frequency gain**

$$A_{v0} = g_{mN} \cdot [(g_{mN} \cdot r_{oN}^2) \parallel (g_{mN} \cdot r_{oP}^2)]$$

- **Open-loop transfer function**

$$A_v(s) = A_{v0} \cdot \frac{1}{(1 + s / \omega_{p1})(1 + s / \omega_{p2})}$$

$$\omega_{p1} \approx \frac{1}{[(g_{mN} \cdot r_{oN}^2) \parallel (g_{mN} \cdot r_{oP}^2)]C_L}$$

- **3-dB bandwidth**

$$BW \approx \frac{\omega_{p1}}{2\pi} \quad (\text{Hz})$$

- **Gain bandwidth product**

$$GBW \approx A_{v0} \cdot BW = \frac{g_{m1,2}}{2\pi \cdot C_L} \quad (\text{Hz})$$

Ex. 9.5: Cascode OTA design example (1)

- $V_{DD} = 3V$; Diff. output swing: 3V; Gain=2000; Power=10mW
 $(\lambda_n=0.1V^{-1}, \lambda_p=0.2V^{-1}; \mu_n C_{ox}=60\mu A/V^2, \mu_p C_{ox}=30\mu A/V^2;$
 $V_{THN}=|V_{THP}|=0.7V; L_{min}=0.5\mu m);$

– Start point: current

- $I_{total}: \approx 3.3mA$

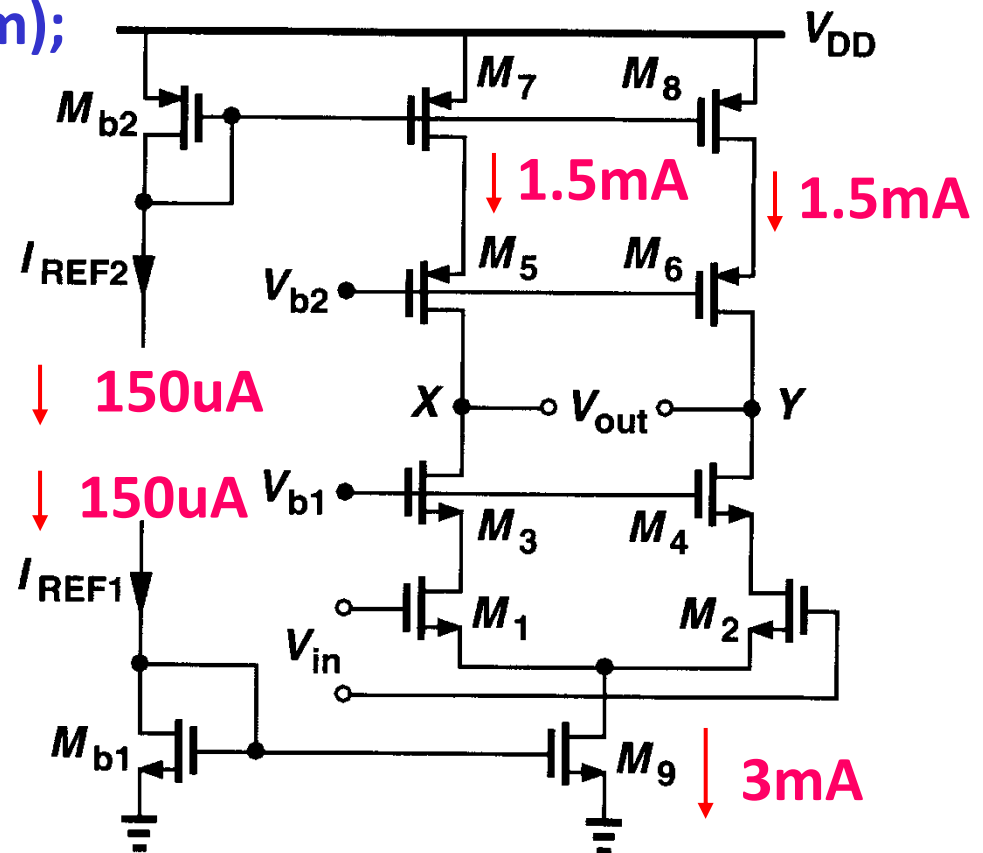
– Output swing

- Allocate V_{OD}

– $V_{OD9} = 0.5$

– $V_{OD1,2,3,4} = 0.2$

– $V_{OD5,6,7,8} = 0.3$



Ex. 9.7: Cascode OTA design example (2)

– Aspect ratios (MOS sizes)

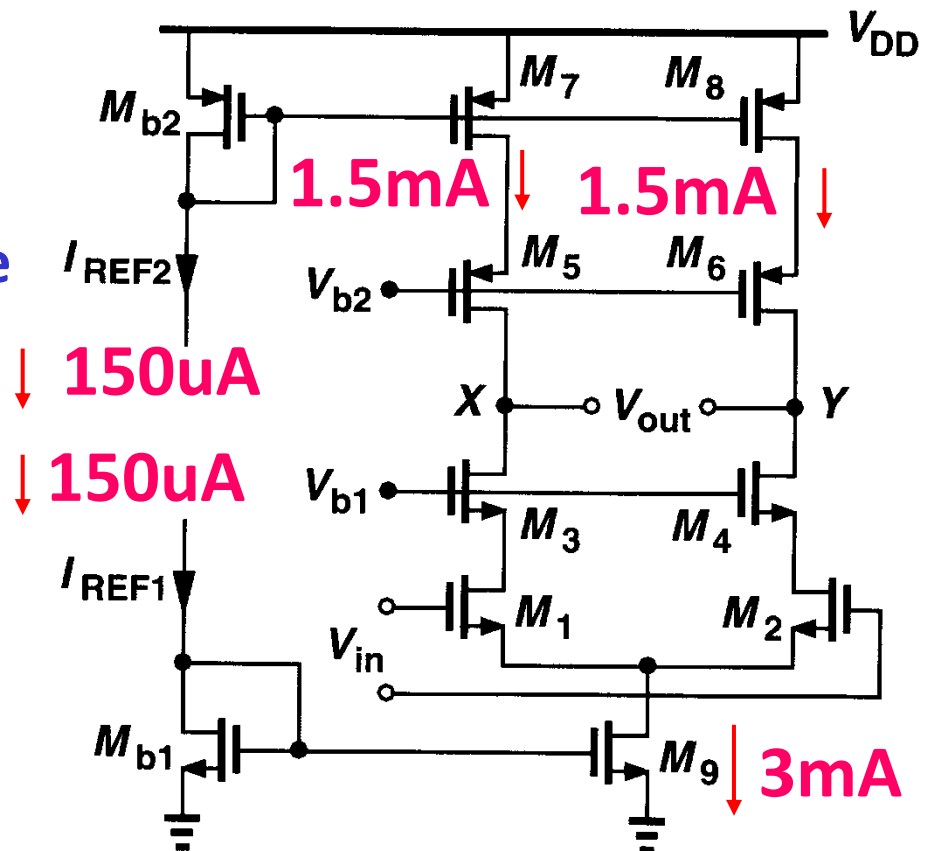
$$I_D = \mu C_{ox} \frac{W}{L} V_{OD}^2$$

– Set L=minimum to reduce size

- $(W/L)_{1-4} = 1250$;
- $(W/L)_{5-8} = 1111$;
- $(W/L)_9 = 400$;
- $(W/L)_{b1} = (1/20)(W/L)_9$;
- $(W/L)_{b2} = (1/10)(W/L)_5$;

– Determine $V_{in,CM}$, V_{b1} , V_{b2} :

- $V_{in,CM} = V_{OD9} + V_{OD1} + V_{THN} = 1.4V$
- $V_{b1} = 0.5 + 0.2 + 0.2 + 0.7 = 1.6V$
- $V_{b2} = 3 - 0.3 - 0.3 - 0.7 = 1.7V$



In practice, body effect must be considered for setting bias voltages (or currents)



Ex. 9.5: Cascode OTA design example (3)

– Calculating gain

$$A_{v0} = g_{m1} [(g_{m3} r_{o3} r_{o1}) \parallel (g_{m5} r_{o5} r_{o7})]$$

$$g_m = \frac{2I_D}{V_{OD}}$$

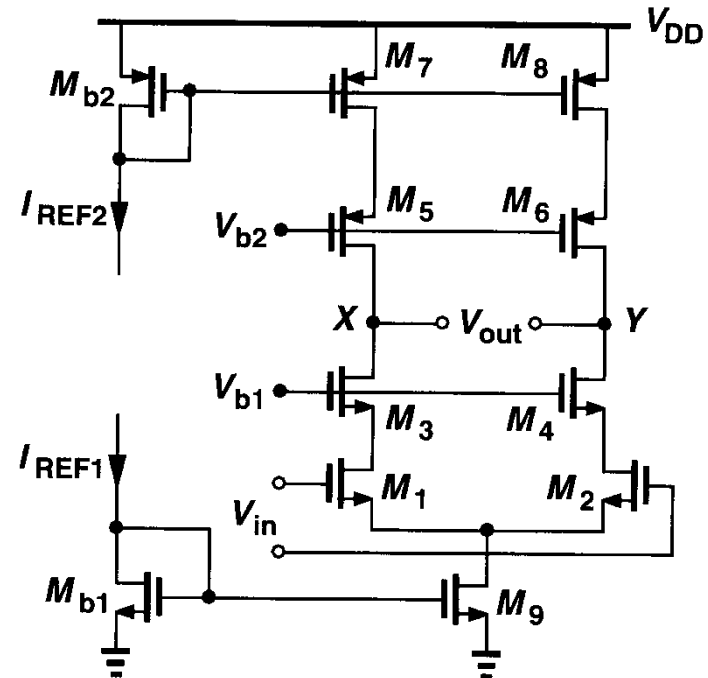
$$r_o = \frac{1}{\lambda I_D}$$

→ $A_{v0} \approx 1250$

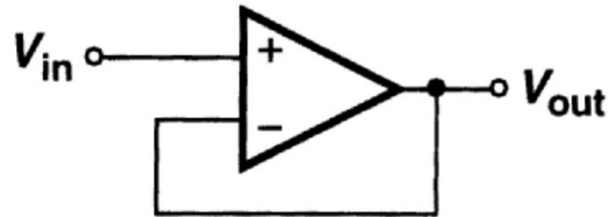
– How to increase gain?

• $(W/L)_{5-8} = 1111 \mu\text{m}/1 \mu\text{m}$

→ $A_{v0} = 4000$



Unity Gain One Stage Cascode

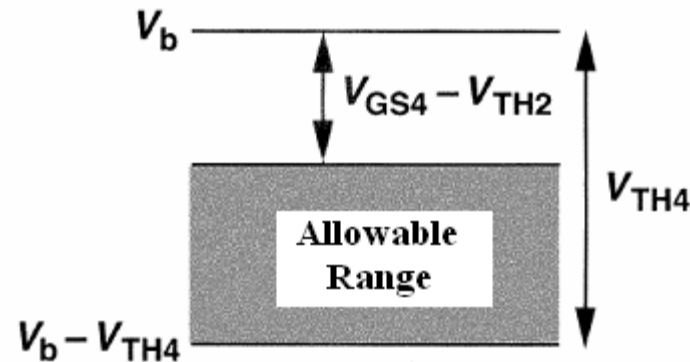
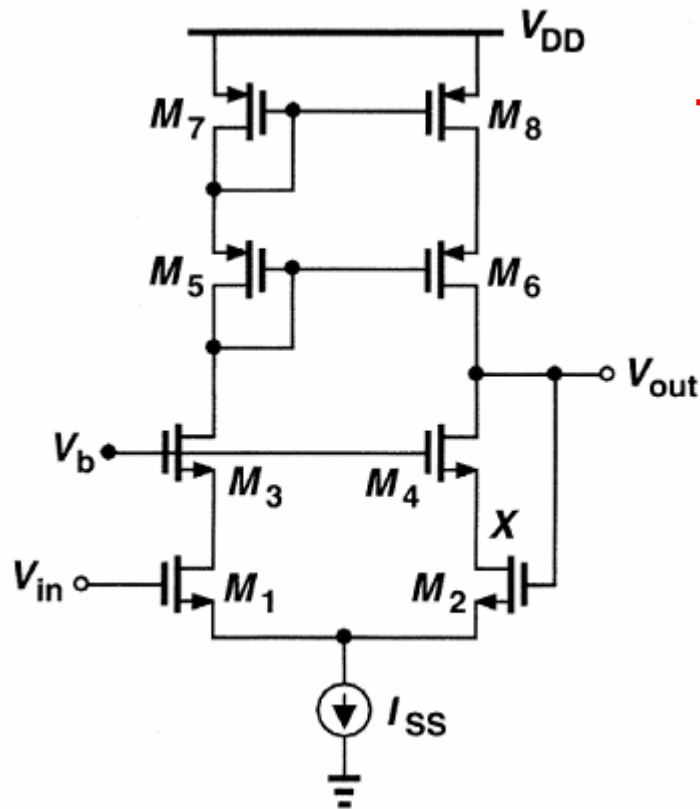


- Connected in unit gain

$$V_b - V_{TH4} \leq V_{out} \leq V_X + V_{TH2}$$

$$V_X = V_b - V_{GS4}$$

$$V_b - V_{TH4} \leq V_{out} \leq V_b - (V_{GS4} - V_{TH2})$$



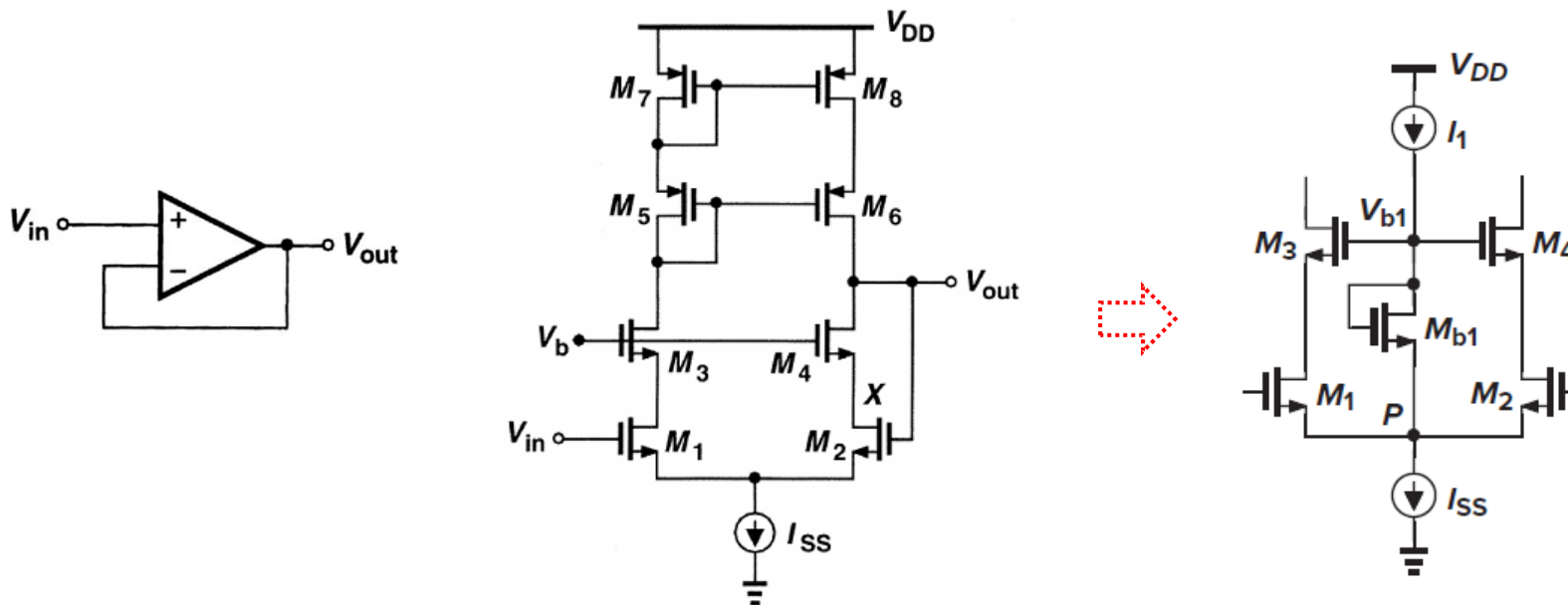
- The output range is limited !!
– Less than one V_{TH}



Improved biasing for cascode

- How to increase the voltage swing of a cascode OTA used in unit-gain feedback?
 - Problem: V_{b1} is fixed, making it difficult to determine the value of V_{CM} .
 - Can V_{b1} tracks V_{CM} ?

$$V_{b1} = V_{in,CM} - V_{GS1,2} + V_{GS,b1}$$



Linear scaling

- Also called power scaling
 - Power budget is different while all other specs remain the same.
- For example: if power is doubled
 - Width of every MOS is doubled (lengths are kept constant)
 - I_D is doubled ; W/L is doubled
 - g_m is doubled
 - V_{OD} is constant
 - r_o is halved
 - As a result: gain, swing are the same while only I_D is scaled linearly.
- Why scaling is needed?

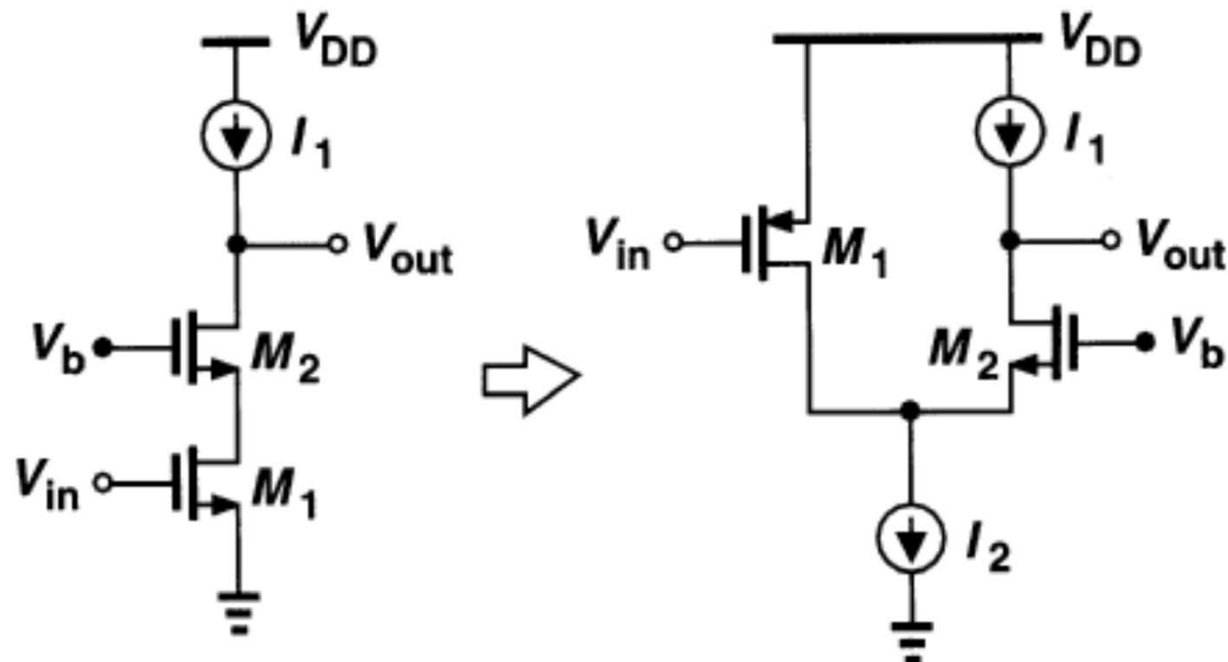


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- **Single-stage OTA**
 - Differential pair
 - Cascode (Telescope) OTA
 - **Folded cascode OTA**
- Two-Stage OTA
- Gain boosting technique
- Common-mode feedback
- Input range limitations
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- Power supply rejection

Folded Cascode Stages (1)

Use folded topology to alleviate the drawbacks of telescopic cascode OTA

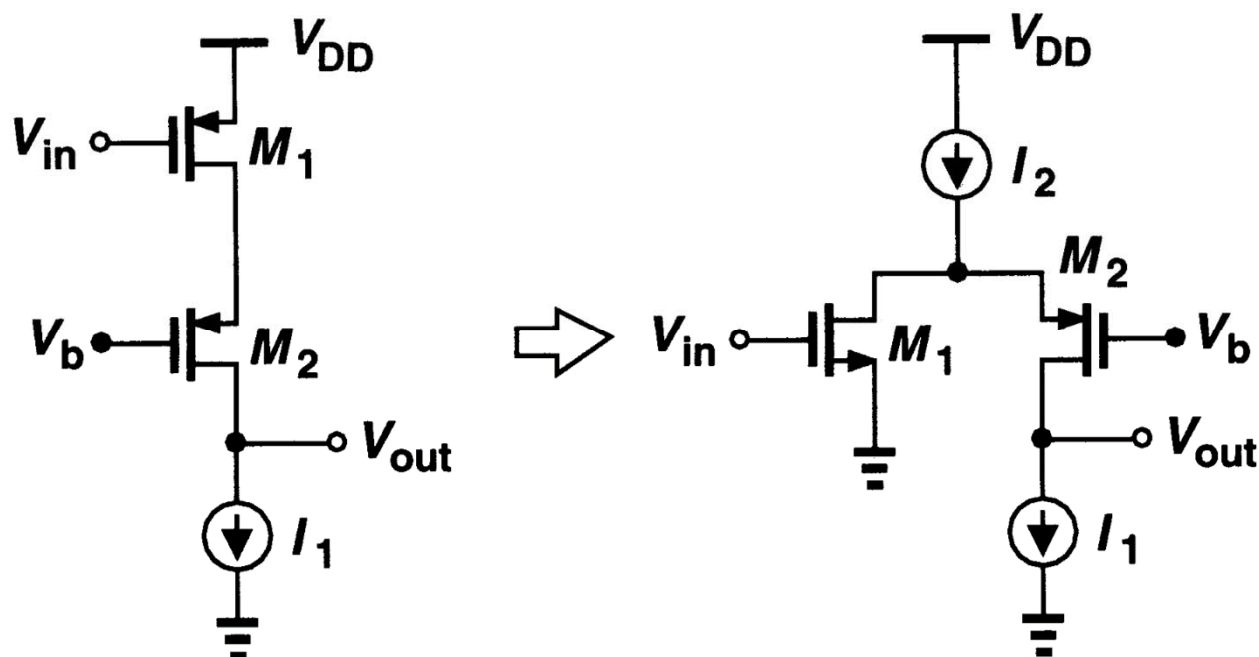


NMOS Cascode

PMOS input

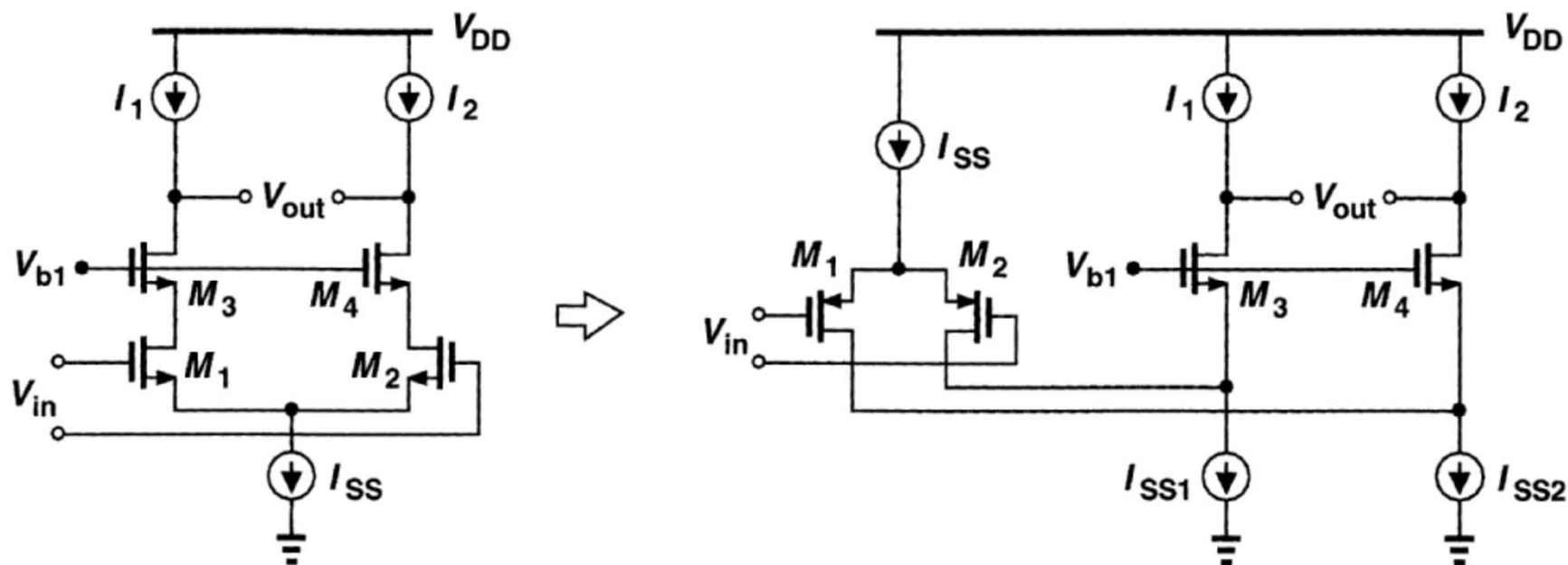


Folded Cascode Stages (2)



PMOS Cascode $\longrightarrow$ NMOS input

Differential Folded Cascode OTA



(1) Bias: Cascode: $I_1 = I_2 = I_{SS}/2$; Folded: $I_{SS1} = I_1 + I_{SS}/2$

(2) Input CM: Cascode: $V_{OD,SS} + V_{GS1} \leq V_{in,CM} \leq V_b - V_{GS3} + V_{TH1}$

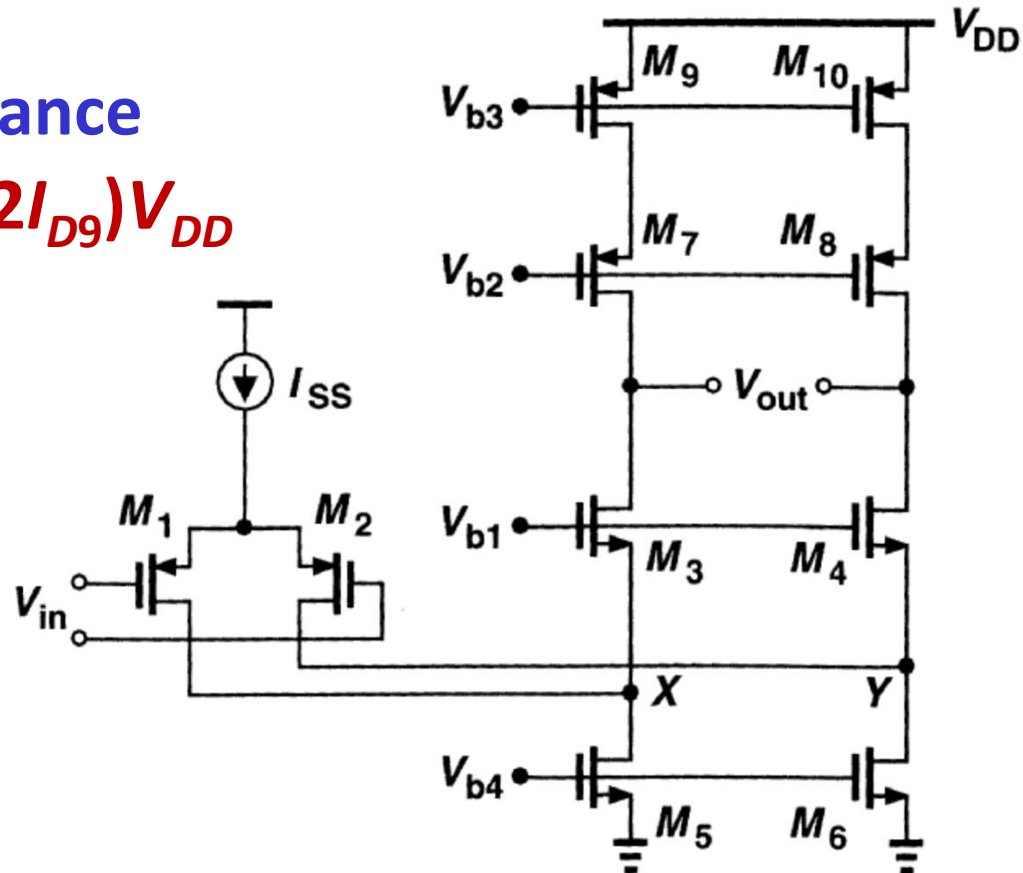
Folded:

$$V_{b1} - V_{GS3} - |V_{TH2}| \leq V_{in,CM} \leq V_{DD} - |V_{OD,SS}| - |V_{GS2}|$$



Full diff. folded Cascode OTA circuit

- Large signal performance
 - Power Diss.: $(I_{SS} + 2I_{D9})V_{DD}$
 - Output DC point
 - Not sure
 - Need CMFB
 - Output Swing

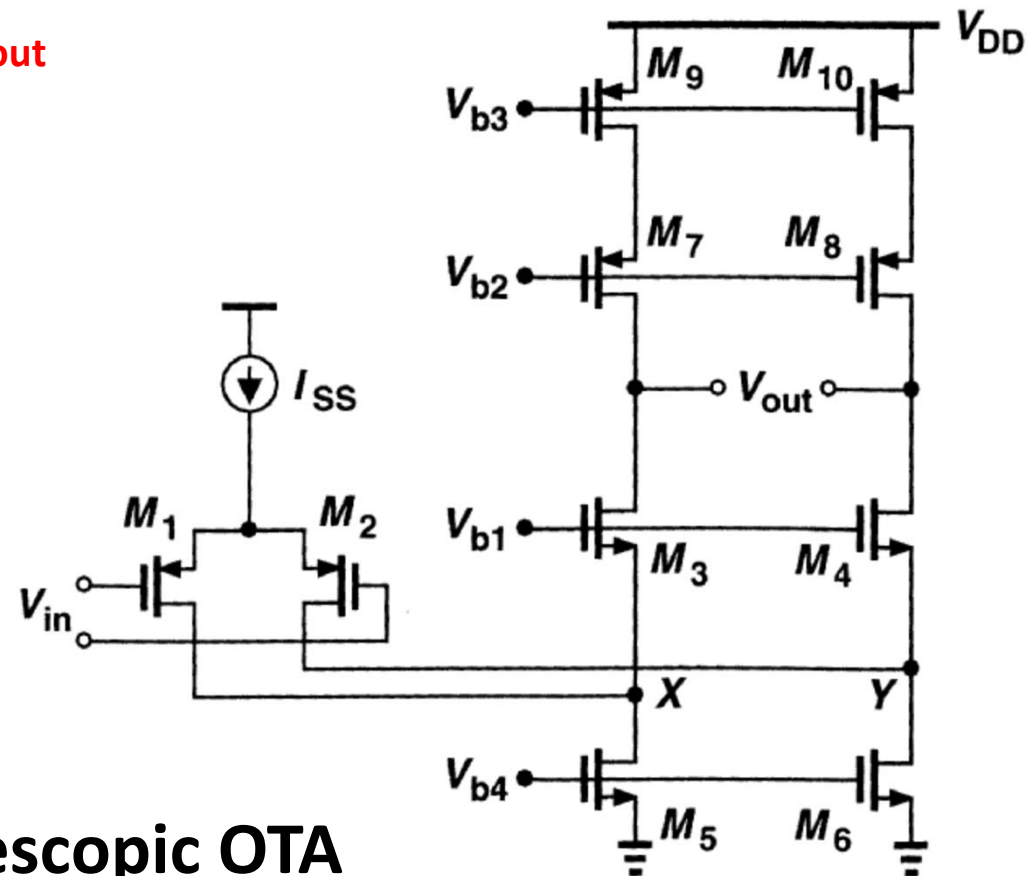
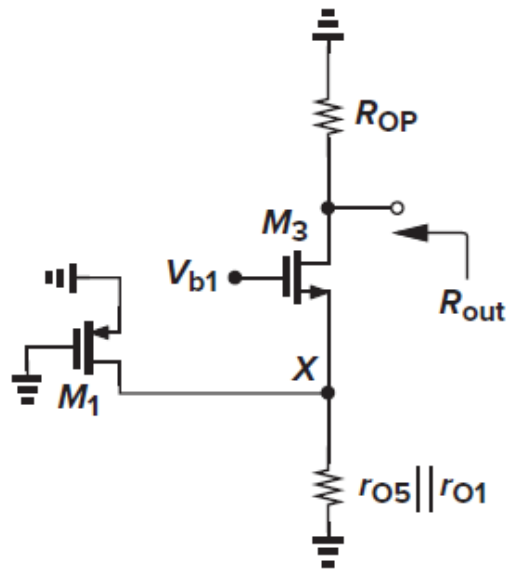


$$V_{OD4} + V_{OD6} \leq V_{out1,2} \leq V_{DD} - |V_{OD8}| - |V_{OD10}|$$



Full diff. folded Cascode OTA circuit

Half-circuit to calculate R_{out}



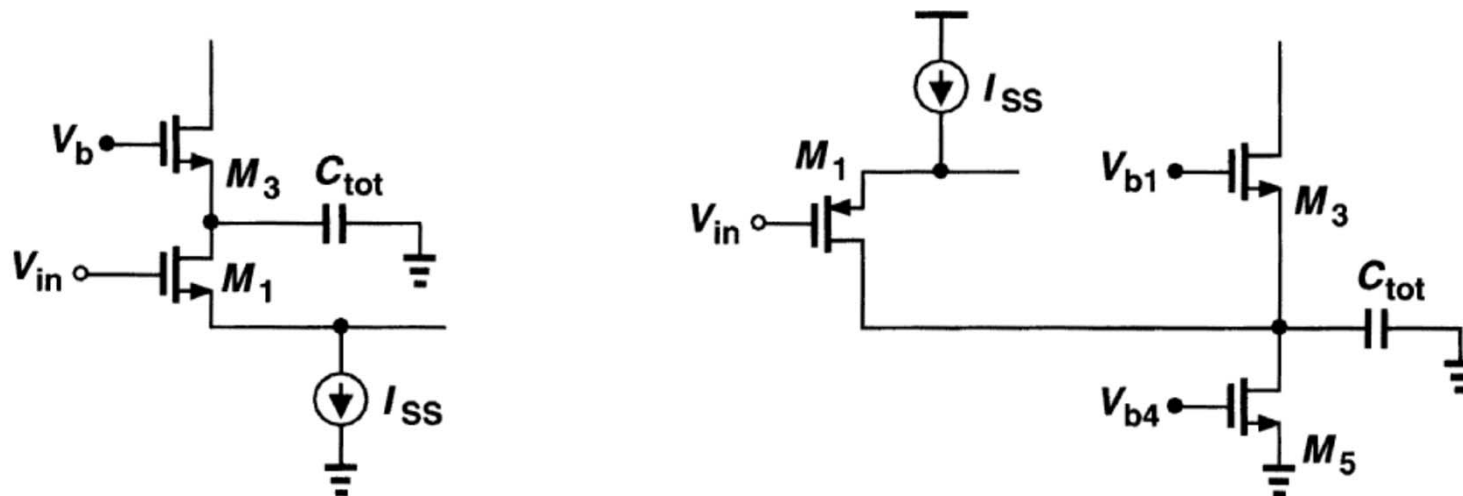
- Small-signal gain
 - A little less than telescopic OTA

$$|A_v| \approx g_{m1} \{ [(g_{m3} + g_{mb3})r_{o3}(r_{o1} \parallel r_{o5})] \parallel [(g_{m7} + g_{mb7})r_{o7}r_{o9}] \}$$



Telescopic vs. Folded Cascode Pole

Pole at the folding point



Telescopic

$$C_{tot} = C_{GS3} + C_{SB3} + C_{DB1} + C_{GD1}$$

Folded:

$$C_{tot} = C_{GS3} + C_{SB3} + C_{DB1} + C_{GD1} + C_{GD5} + C_{DB5}$$

Larger cap. → Lower non-dominant pole freq.





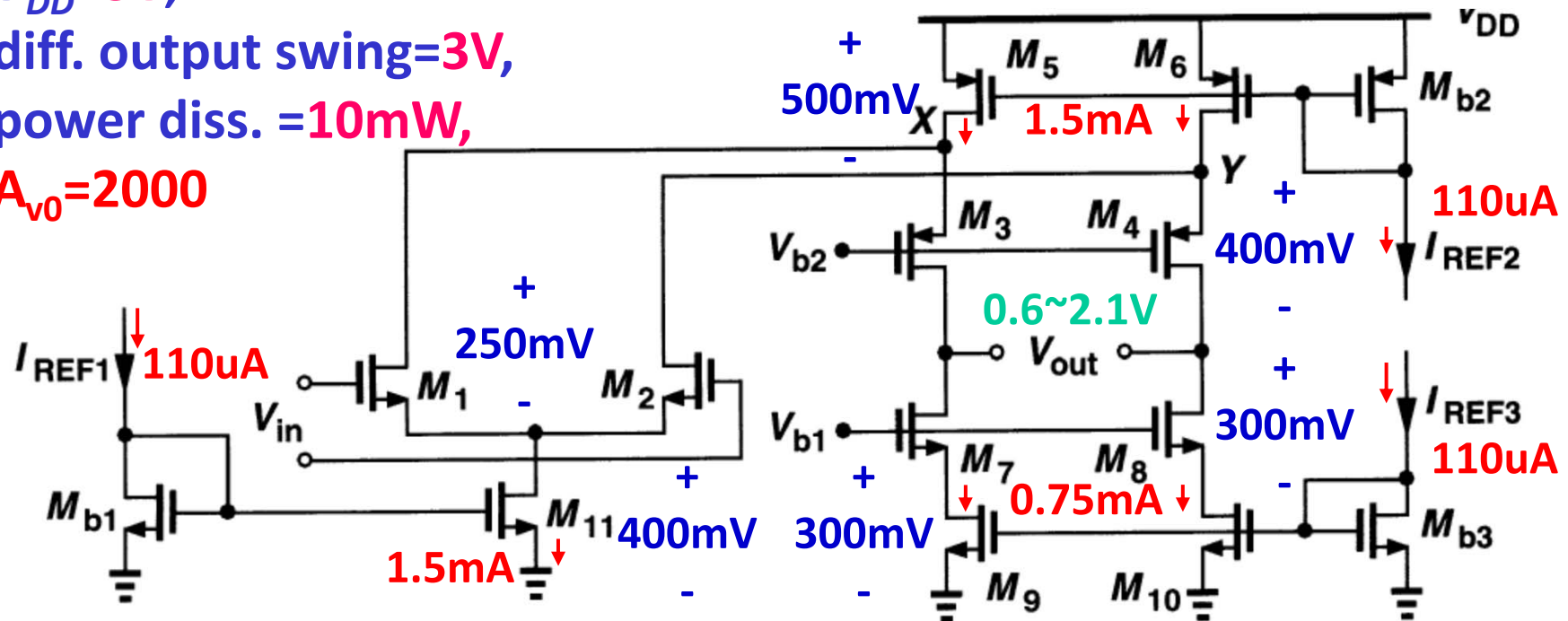
Example 9.9 (1)

$V_{DD}=3V$;

diff. output swing=3V,

power diss. =10mW,

$A_{v0}=2000$



- Start point: total current = $10mW/3V \approx 3.3mA$
- Next: allocate V_{OD} based on the required output swing
 - To keep medium size, larger V_{OD} for larger current
 - $V_{in,CM} = V_{out,CM} = 1.35 \rightarrow V_{OD1} = V_{OD2} = 0.25V$



Example 9.6 (3)

$$|A_v| \approx g_{m1} \{ [(g_{m3} r_{o3} \cdot (r_{o1} \parallel r_{o5})) \parallel (g_{m7} r_{o7} \cdot r_{o9})] \}$$

$$g_m = \frac{2I_D}{V_{OD}} \quad r_o = \frac{1}{\lambda I_D} \quad \begin{array}{l} \lambda_n = 0.1\text{V}^{-1}, \\ \lambda_p = 0.2\text{V}^{-1} \\ L_{min} = 0.5\text{mm} \end{array}$$

$$\rightarrow g_{m1,2} = 0.006\text{A/V}; \quad g_{m3,4} = 0.0038\text{A/V}; \quad g_{m7,8} = 0.005\text{A/V}$$

$$\rightarrow r_{o1,2} = r_{o7-10} = 13.3\text{K}\Omega, \quad r_{o3,4} = 2r_{o5,6} = 6.67\text{K}\Omega$$

$$\longrightarrow A_{v0} \approx 400$$

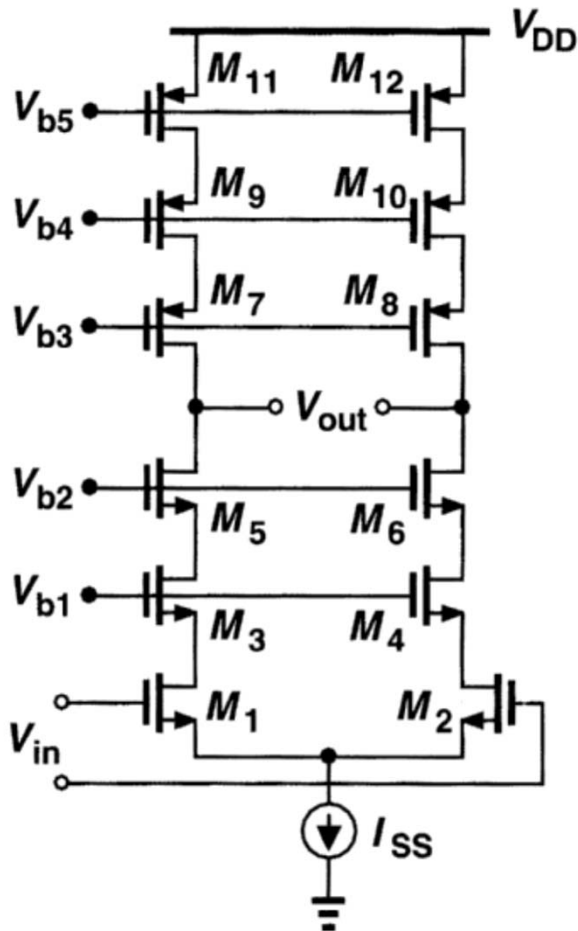
Methods for increasing the gain: (verify them by yourself)

$\rightarrow L_{5,6}$ can be increased to enhance r_{o5}/r_{o6} ;

$\rightarrow g_{m1,2}$ can be increased by widen $W_{1,2}$.



Triple Cascode



$$|A_v| \approx (g_m r_o)^3 / 2$$

High Gain, but:

Limited Output Swing

Complex biasing



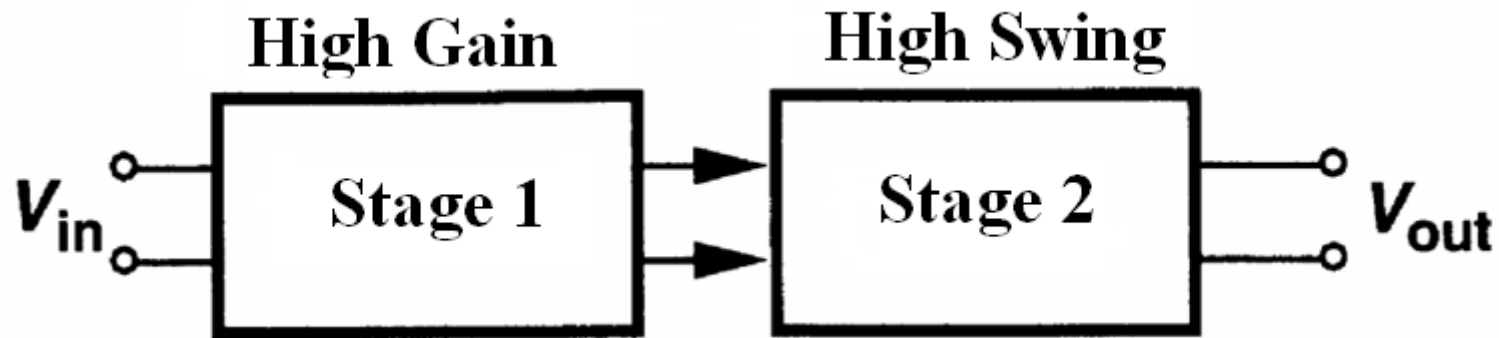
Outline

- General consideration
- Single-stage OTA
- **Two-Stage OTA**
- Gain boosting technique
- Common-mode feedback
- Input range limitations
- Slew Rate
- Power supply rejection

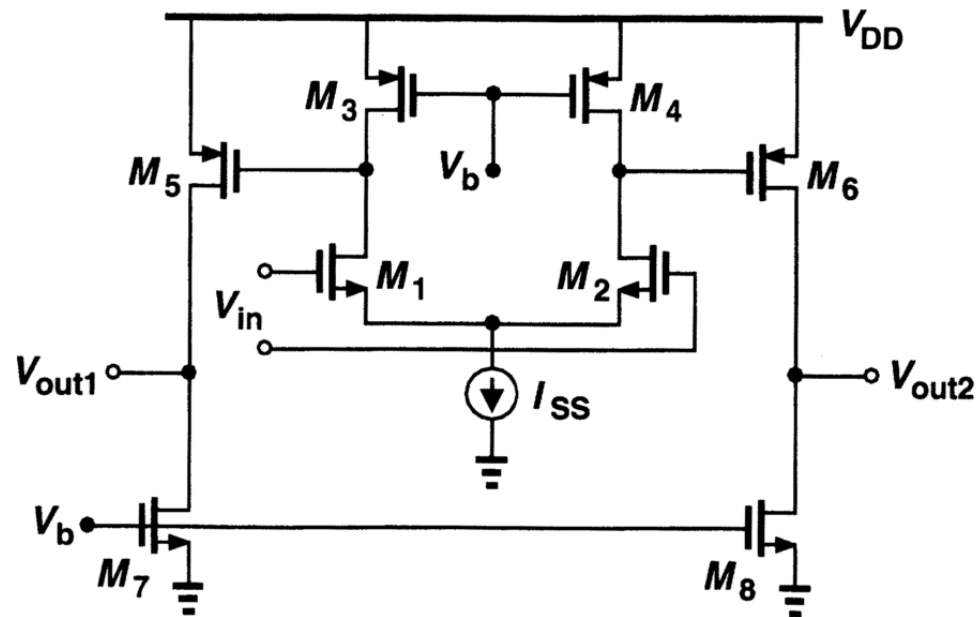


Conceptual model of two-stage OTA

Design Approach for Two-Stage Op Amps



Two-Stage Op Amp



$$A_v = [g_{m1}(r_{O1} \parallel r_{O3})] \times [g_{m5}(r_{O5} \parallel r_{O7})]$$

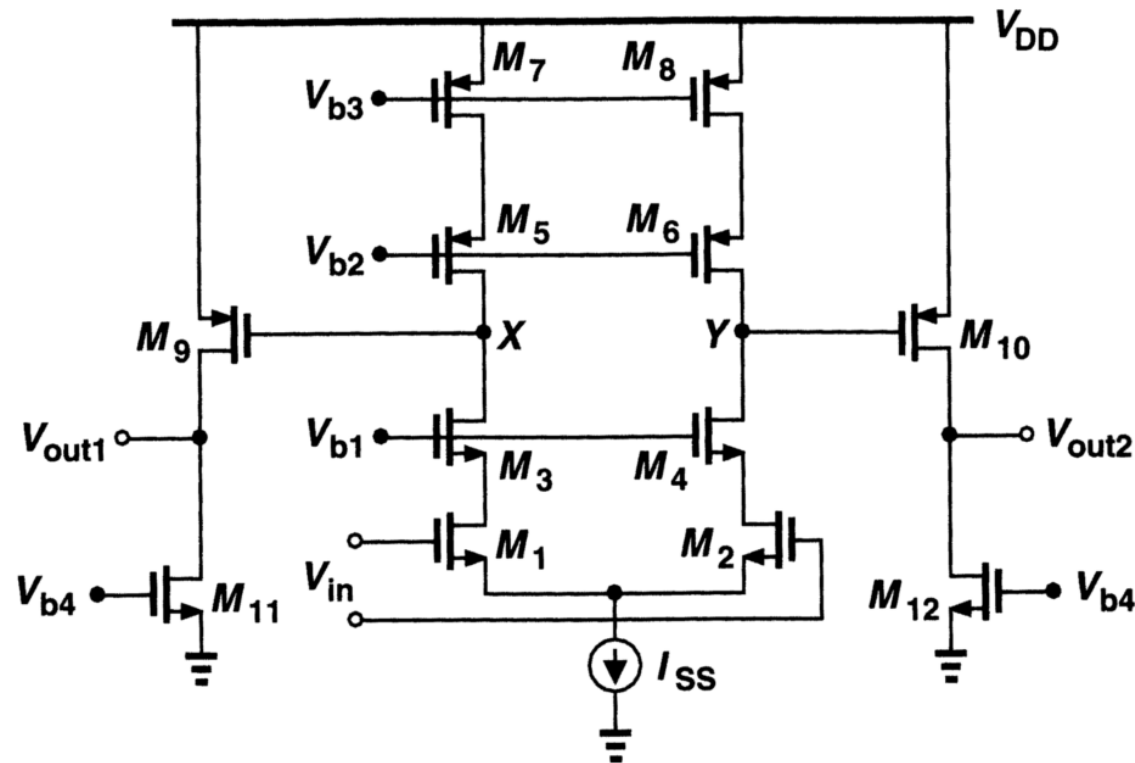
Pros: very high gain and swing, with medium power

Cons: (1) stability issues

(2) difficulties in setting OP for the second stage

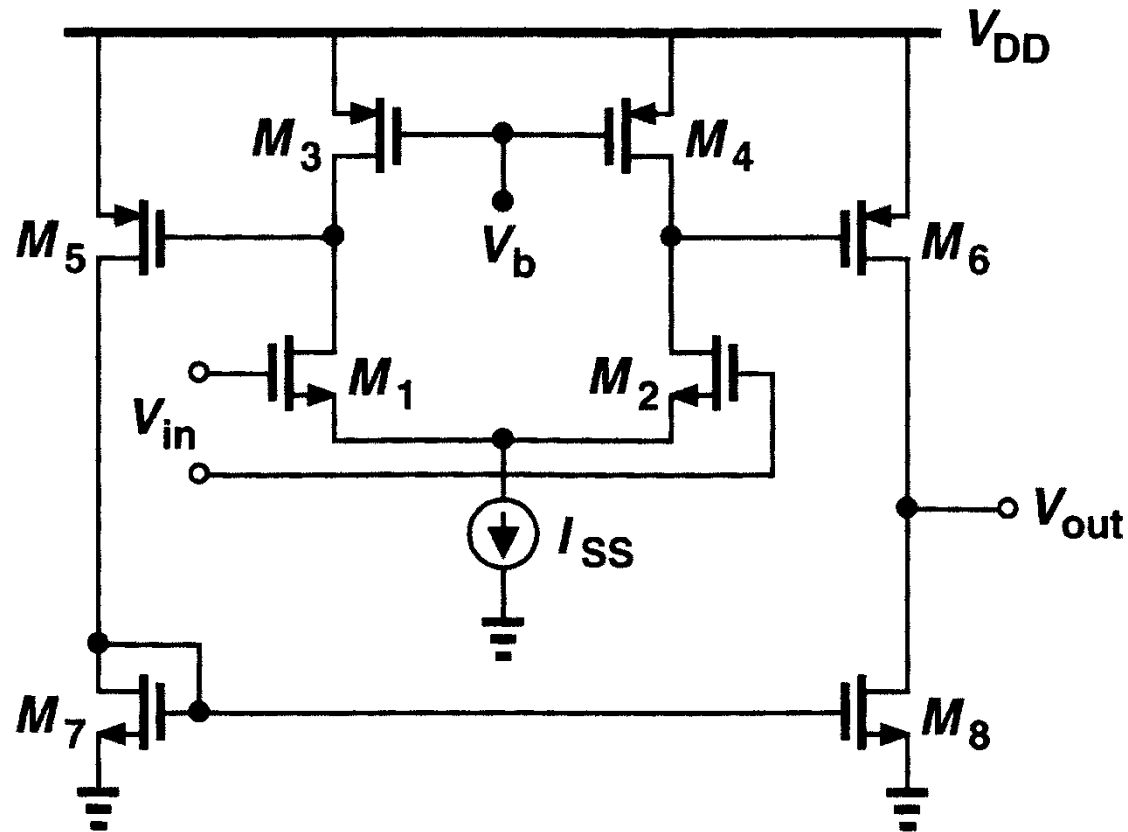


Two-Stage OTA with very high gain



$$A_v = \{g_{m1}[(g_{m3}r_{O3}r_{O1}) \parallel (g_{m5}r_{O5}r_{O7})]\} \times [g_{m9}(r_{O9} \parallel r_{O11})]$$

Two-stage opamp with single-ended output

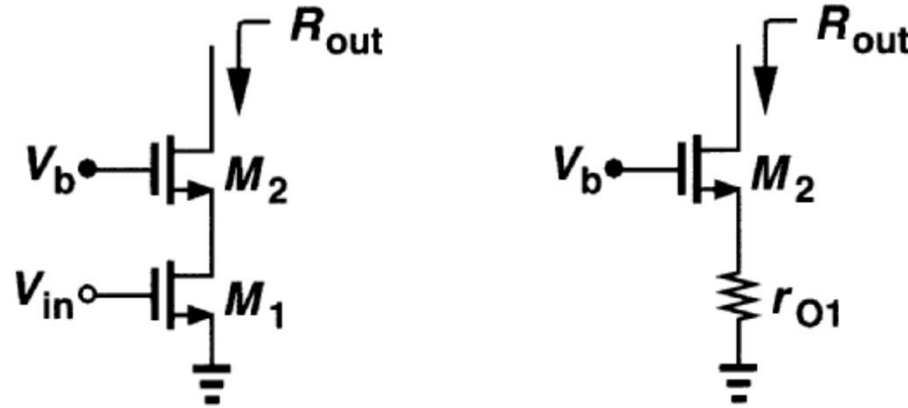


Outline

- General consideration
- Single-stage OTA
- Two-Stage OTA
- **Gain boosting techniques**
- Common-mode feedback
- Input range limitations
- Slew Rate
- Power supply rejection

Enhance the output impedance of Cascode

Cascode



$$R_{out} = g_{m2} r_{O2} r_{O1}$$

How to increase the output impedance further?

-- Fall back on feedback!!

What kind of feedback we need?



Output Impedance Enhancement With Feedback

Regulated Cascode

Essentially a current voltage feedback!

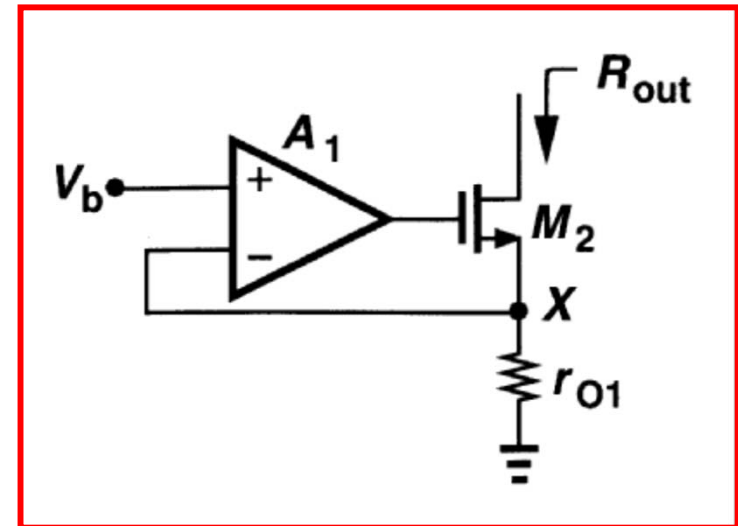
Loop gain

the path from gate of M_2 to X forms a source follower

→ Loop gain $\approx A_1$

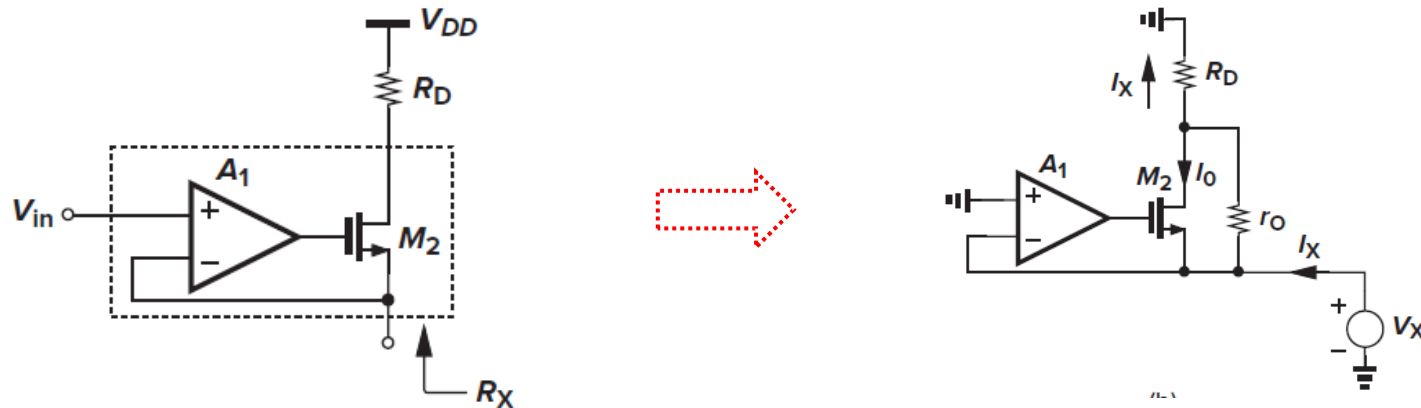
$$R_{out} = R_{out,open} \cdot (1 + A_1)$$

→ $R_{out} \approx A_1 g_{m2} r_{O2} r_{O1}$



Ex. 9.11

Determine the resistance seen at the source of M2 ($\gamma=0$)



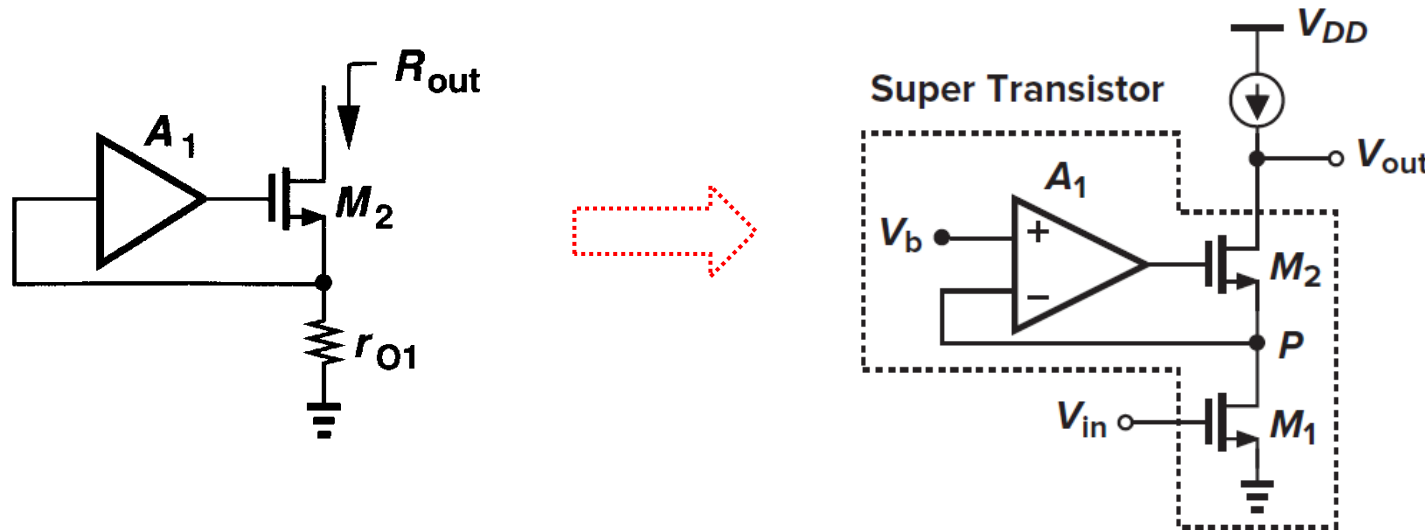
$$\begin{cases} i_x = \frac{v_x - v_{out}}{r_o} - [(0 - v_x)A_1 - v_x]g_m \\ v_{out} = i_x R_D \end{cases} \Rightarrow R_X = \frac{R_D + r_o}{1 + (A_1 + 1)g_m r_o}$$

For $R_D \ll r_o$, $R_X \approx \frac{1}{(A_1 + 1)g_m}$

Without A_1 , $R_X \approx 1/g_m$



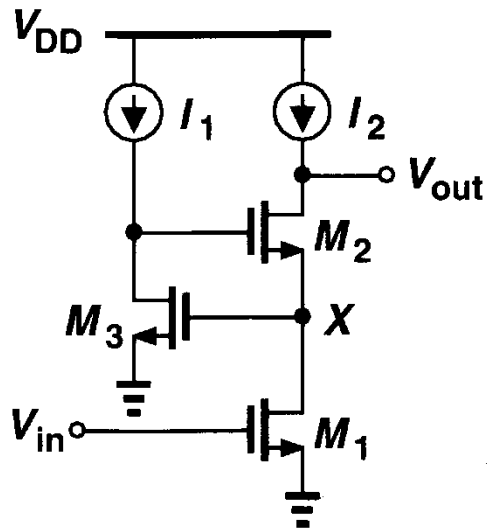
Gain Boosting in cascode Stage



- For the super transistor
 - $G_m \approx g_{m1}$ (why? See derivation of (9.34) for details)
 - $R_{out} \approx A_1 g_{m2} r_{o2} r_{o1}$
 - $V_{out,min} = V_{OD1} + V_{OD2}$ (with a proper A_1)
- Comparing with a cascode:
 - Gain is boosted while voltage swing remains the same



Circuit implementation



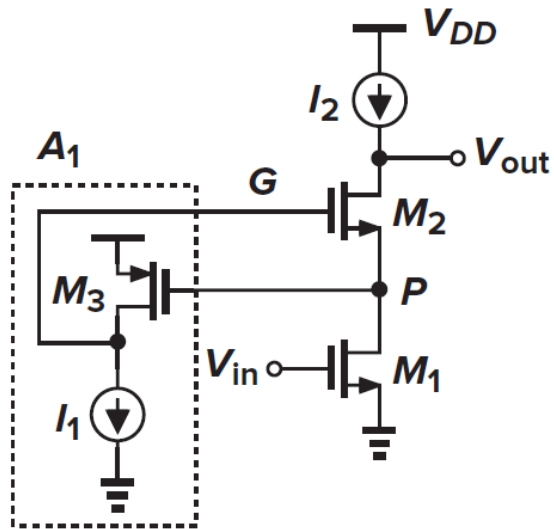
$$|A_V| \approx g_{m1} (g_{m2} r_{O2} r_{O1}) (g_{m3} r_{O3})$$

$$V_X = V_{GS3}$$

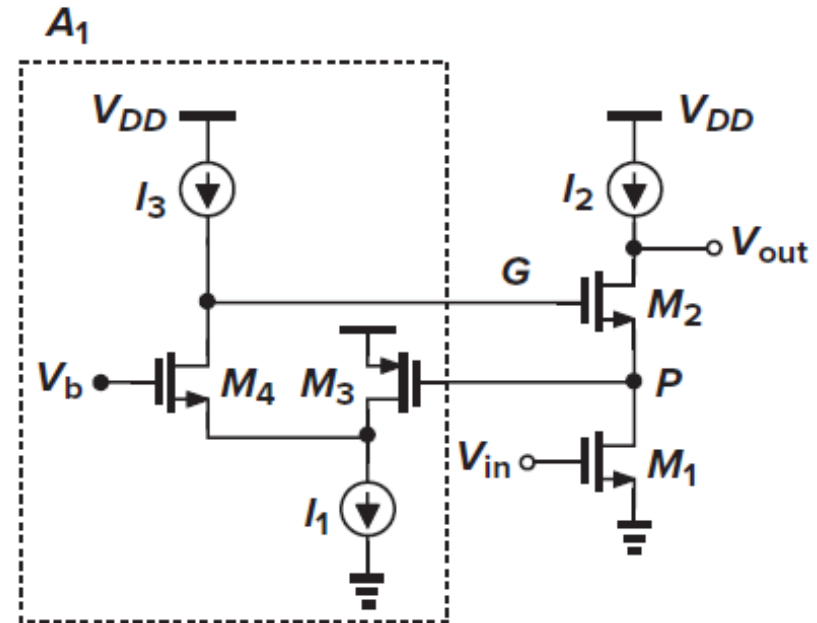
- Gain is boosted, but
 - the output range is reduced
 - nearly by one V_{TH} !!



Solutions to improve the swing



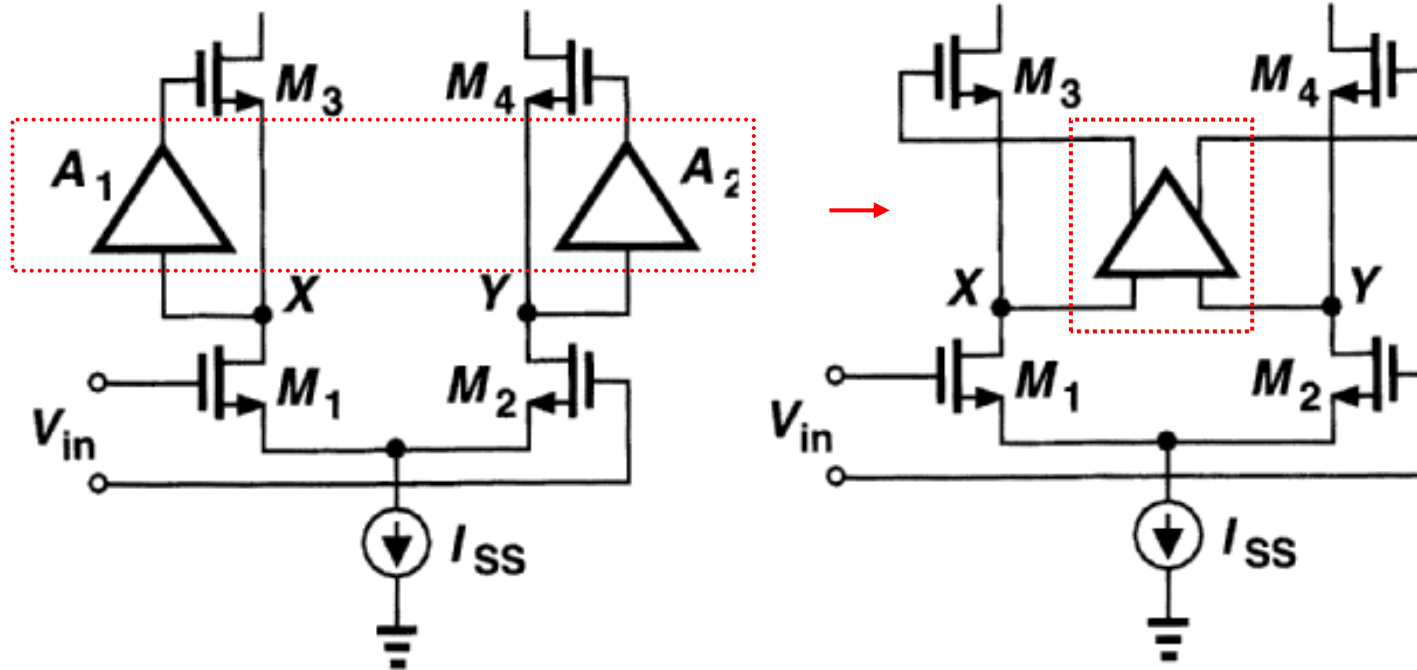
- CS with PMOS input



- Folded Cascode with PMOS input



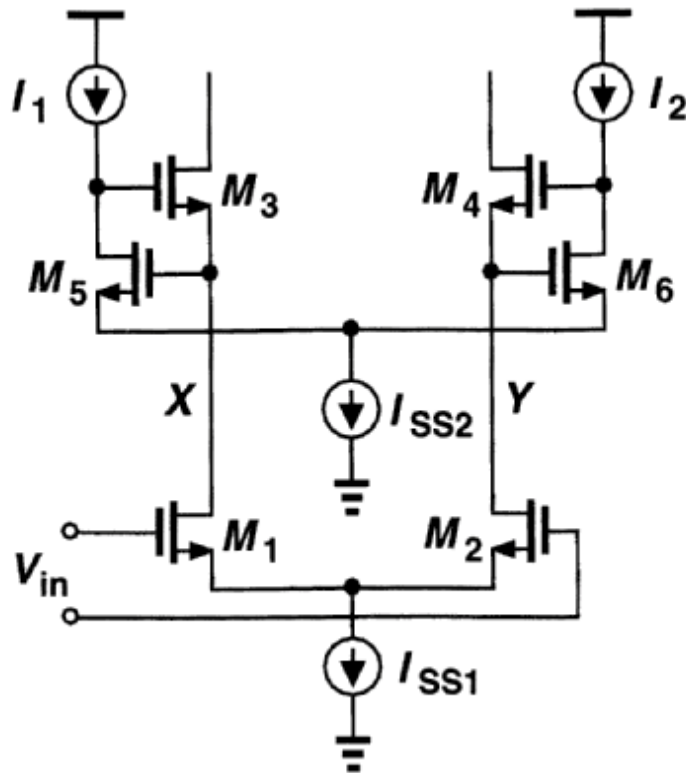
Differential Gain Boosting (1)



Boosting the impedance of a differential cascode stage



Differential Gain Boosting (2)



The minimum level at the drain of M_3 becomes:

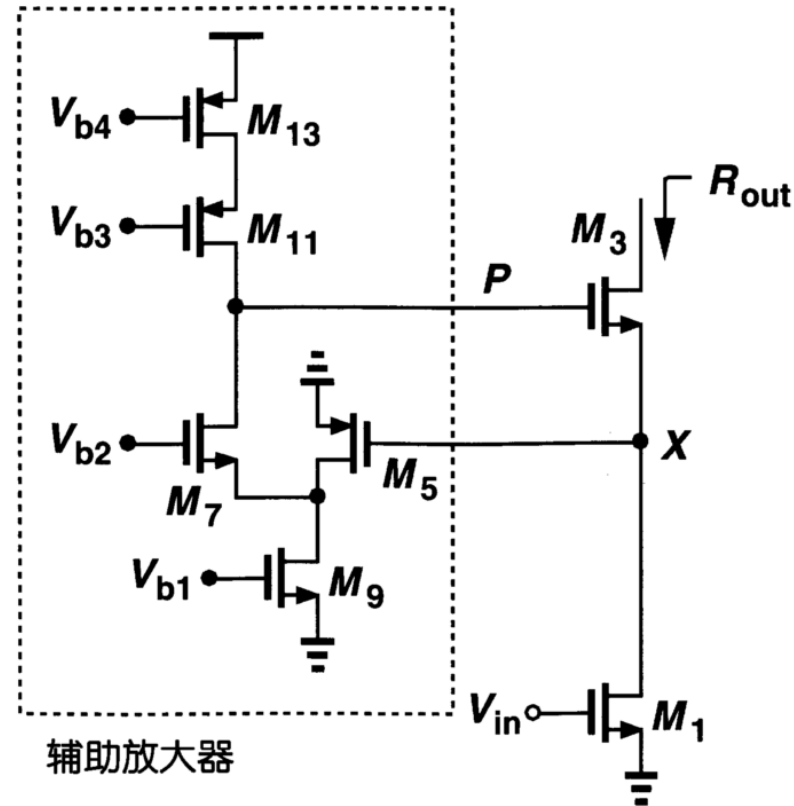
$$V_{OD3} + V_{GS5} + V_{OD,ISS2}$$

This level would be one threshold voltage higher.





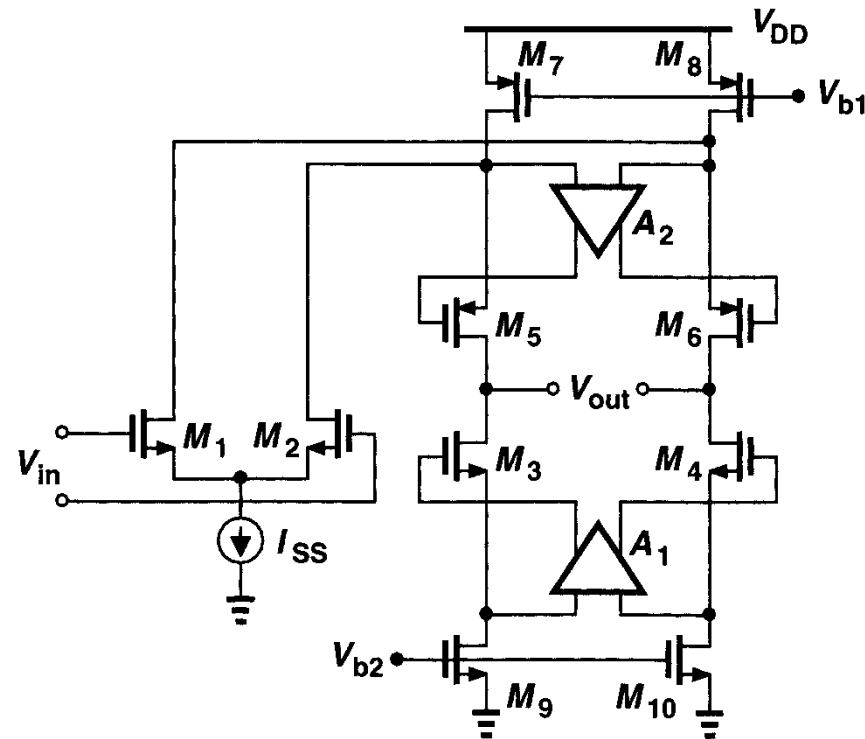
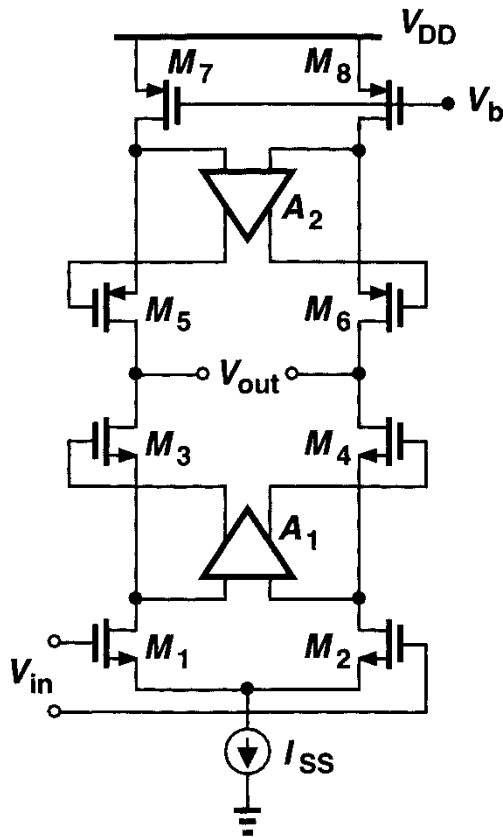
Example 9.14 Calculate the output impedance



$$R_{out} \approx \underbrace{g_{m3} r_{O3} r_{O1}}_{\text{red line}} \underbrace{g_{m5} \{ [g_{m7} r_{O7} (r_{O9} \parallel r_{O5})] \parallel (g_{m11} r_{O11} r_{O13}) \}}_{\text{green line}}$$



Practical gain boosted OTA

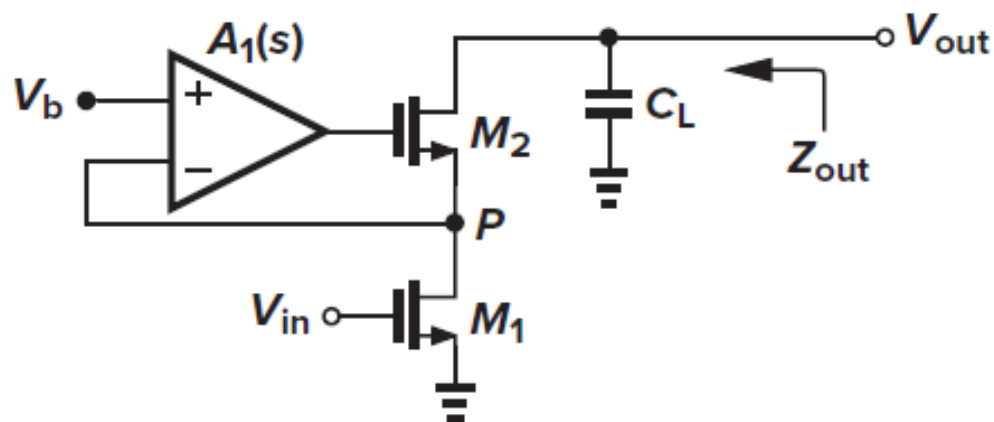


Gain boosting applied to both signal path and load devices.

→ Widely used in high-resolution and high-speed ADCs



Frequency response



- Main OP

- Dominant pole frequency:
 - $1/(A_0 g_{m2} r_{o2} r_{o1} C_L)$
- Non-dominant pole frequency
 - $A_0 g_{m2} / C_p$

- Aux. OP

- Assuming a first-order $A(s)$ with gain of A_0 and pole frequency of ω_0
- What will the overall frequency response look like ?
- How to determine $A_0 \omega_0$?
- See the original paper: “Klaas Bult, et al. The CMOS Gain Boosting Technique, Analog Integrated Circuits and Signal Processing, 1991.”

Comparison of different OTA topologies

	Gain	Output Swing	Speed	Power Dissipation	Noise
Telescopic	Medium	Medium	Highest	Low	Low
Folded-Cascode	Medium	Medium	High	Medium	Medium
Two-Stage	High	Highest	Low	Medium	Low
Gain-Boosted	High	Medium	Medium	High	Medium



Outline

- General consideration
- Single-stage OTA
- Two-Stage OTA
- Gain boosting technique
- **Common-mode feedback**
- Input range limitations
- Slew Rate
- Power supply rejection



Benefits and drawbacks of fully-diff. OTA

- **Benefits**

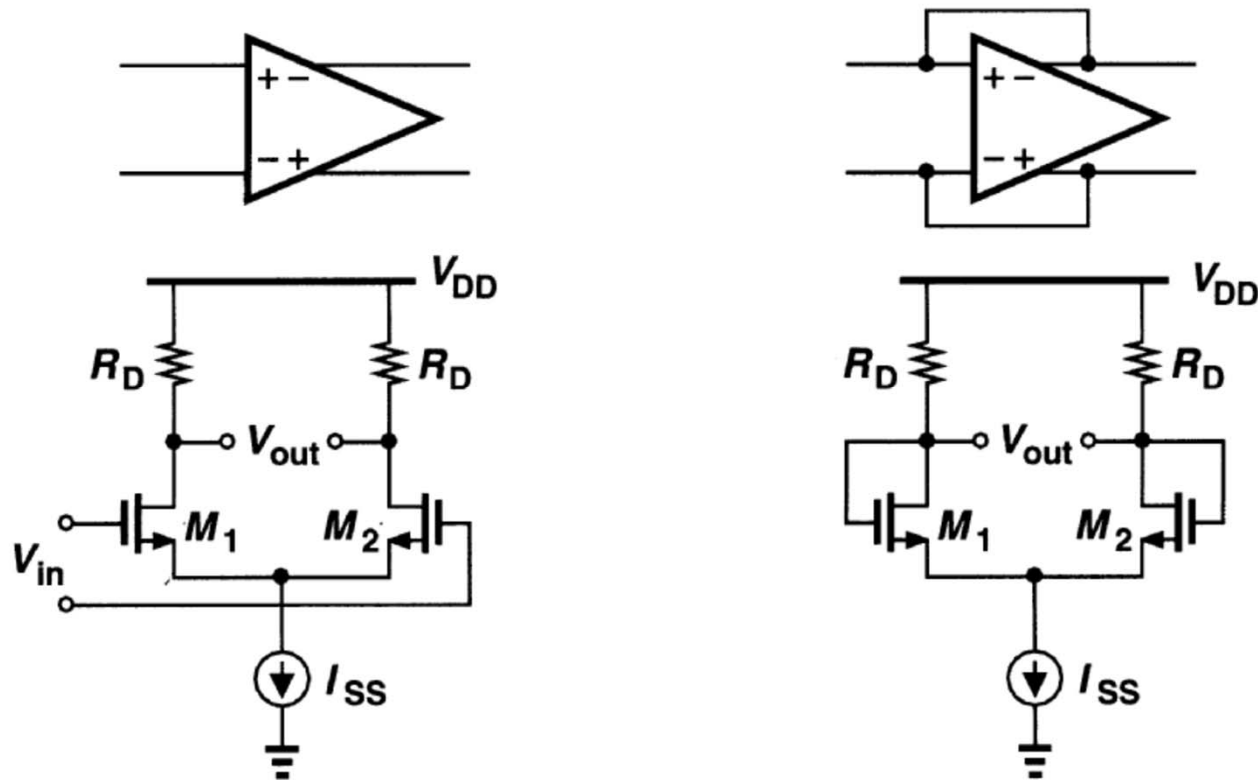
- CM noise and disturbance rejection
- Larger output swing
- no mirror pole and zero → higher speed

- **Drawback**

- the output common-level is not sure
- Common-mode feedback (CMFB) is needed



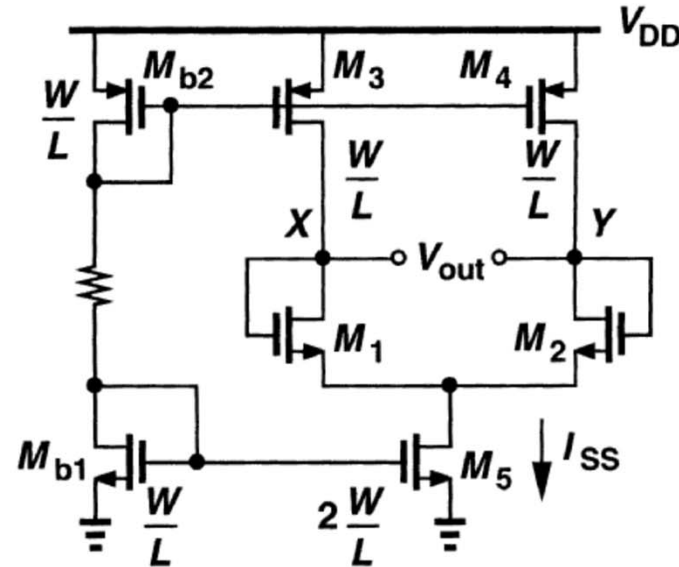
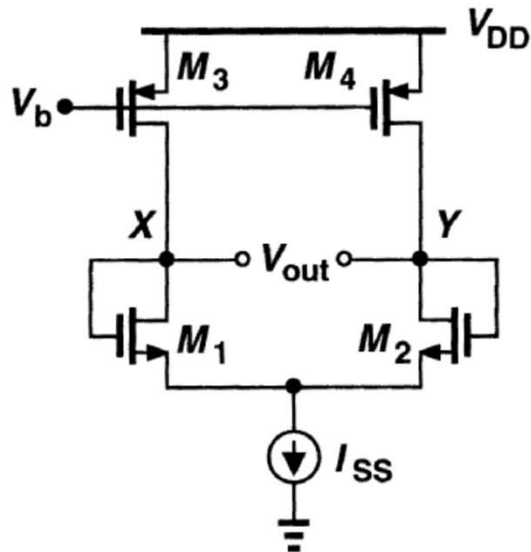
CMFB is not needed for resistive load



$$V_{CM} = V_{DD} - (I_{SS}/2)R_D$$



CMFB must adopted in current source load



$$I_{D3,4} > I_{SS}/2$$

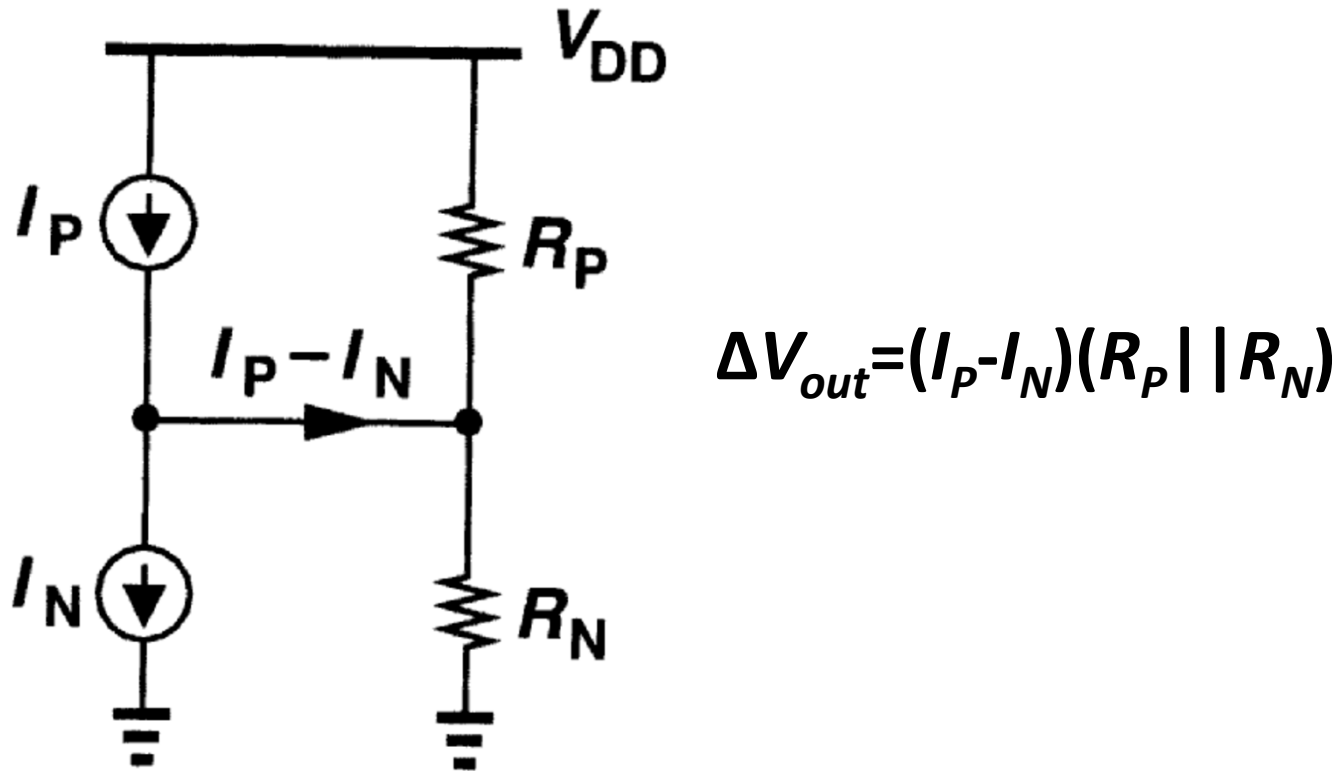
$\Rightarrow V_{X,Y}$ increases $\Rightarrow M_{3,4}$ enter the triode region

$$I_{D3,4} < I_{SS}/2$$

$\Rightarrow V_{X,Y}$ decreases $\Rightarrow M_5$ enter the triode region



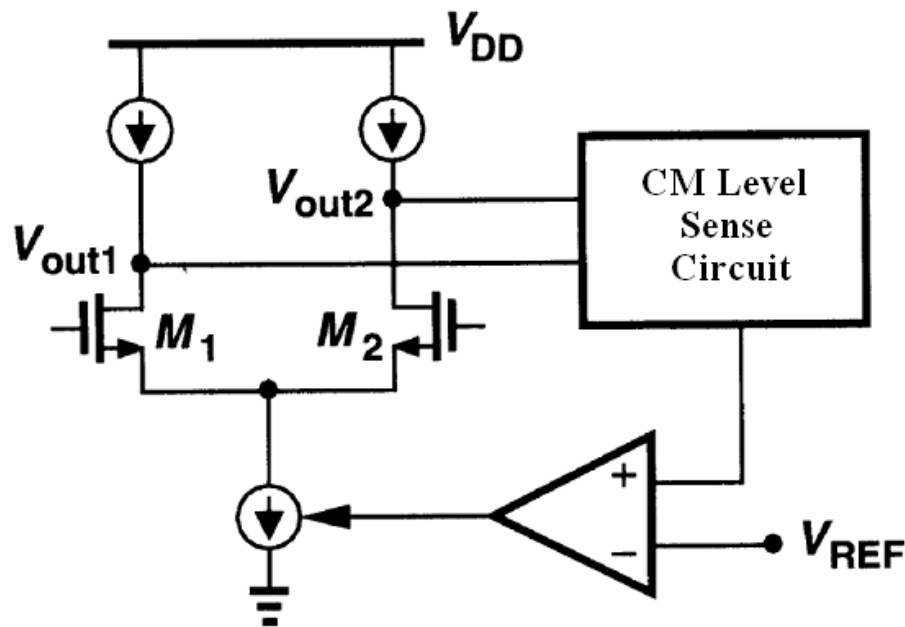
High Gain Amp Model



Simplified model of high-gain amplifier



Conceptual model for CMFB

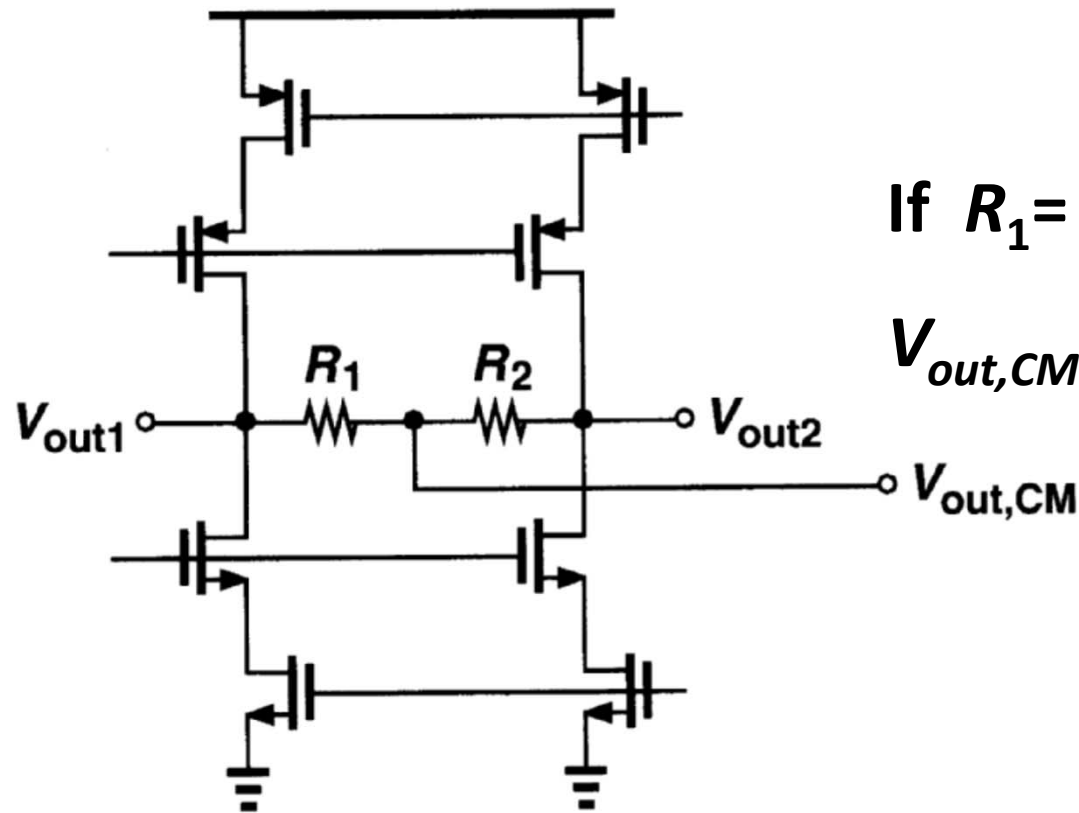


CMFB's 3 operations

1. Sensing the $V_{out,CM}$;
2. Comparison with V_{ref}
3. Returning the error to the amplifier's bias network



$V_{out,CM}$ sensing I: Resistive



$$\text{If } R_1 = R_2 = R \gg R_{out1,2}$$

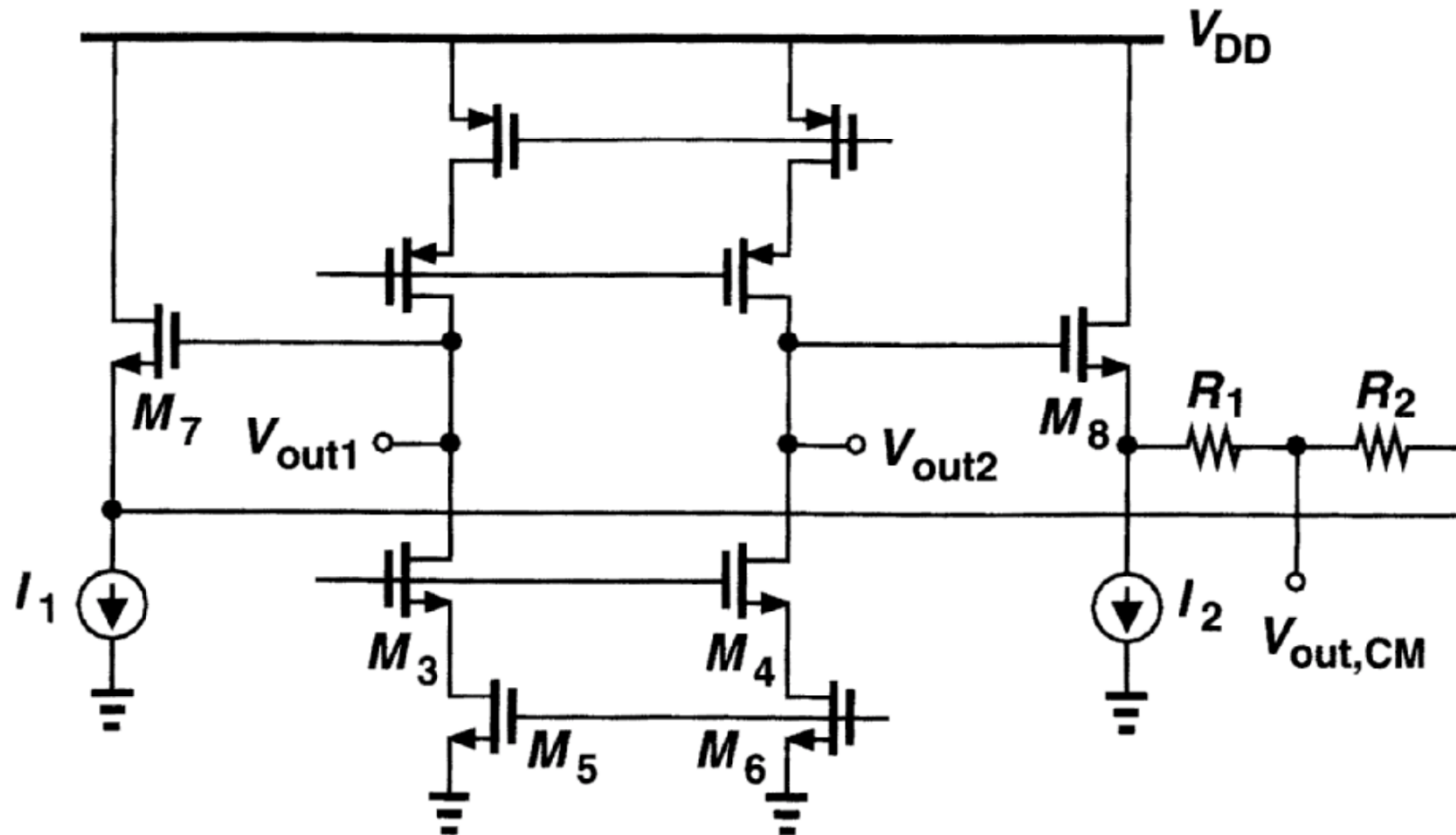
$$V_{out,CM} = (V_{out1} + V_{out2})/2$$

Disadvantage:

very large resistor $\rightarrow$ large die area.



$V_{out,CM}$ sensing II: source-follower

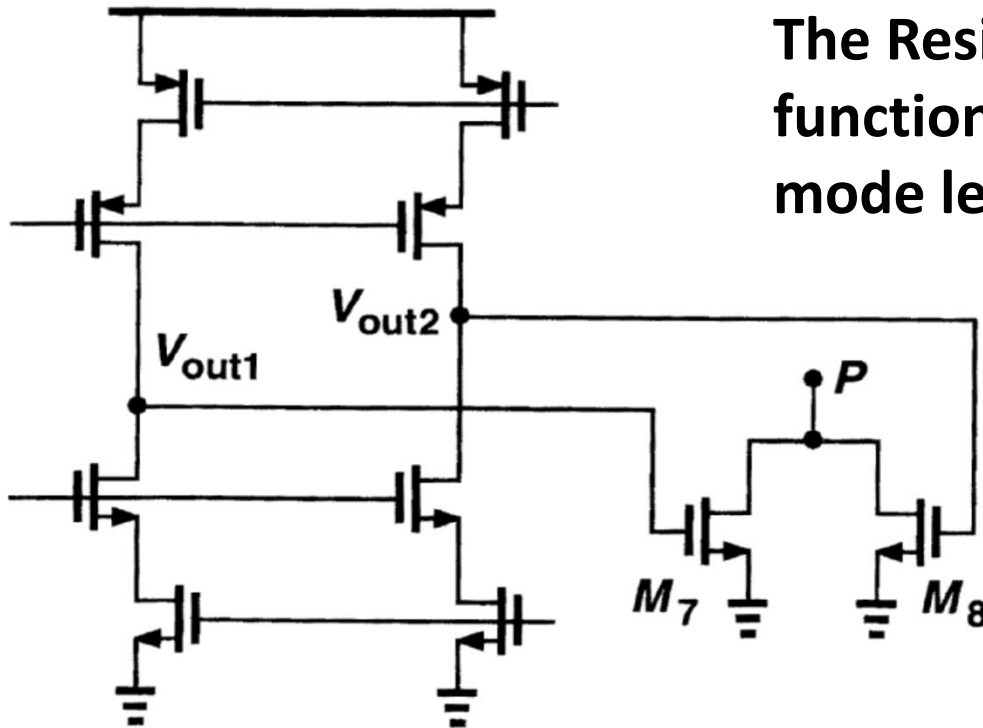


Use source follower to avoid very large resistors

$$V_{out,CM} = (V_{out1} + V_{out2})/2 - V_{GS7,8}$$



$V_{out,CM}$ sensing III: Deep Triode MOS

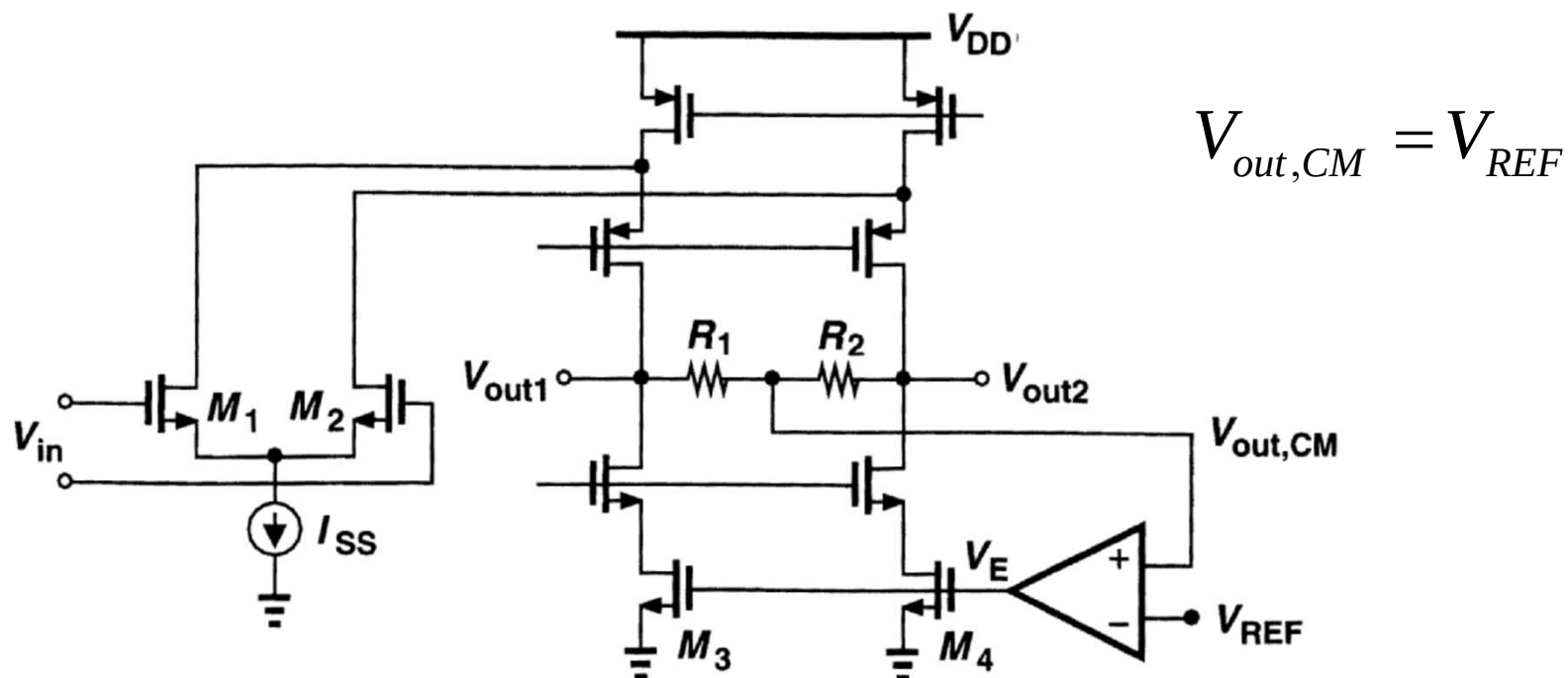


The Resistance at point P is a function of the output Common mode level

$$R_{tot} = R_{on7} \parallel R_{on8} = \frac{1}{\mu_0 C_{OX} \frac{W}{L} (V_{out2} + V_{out1} - 2V_{TH})}$$



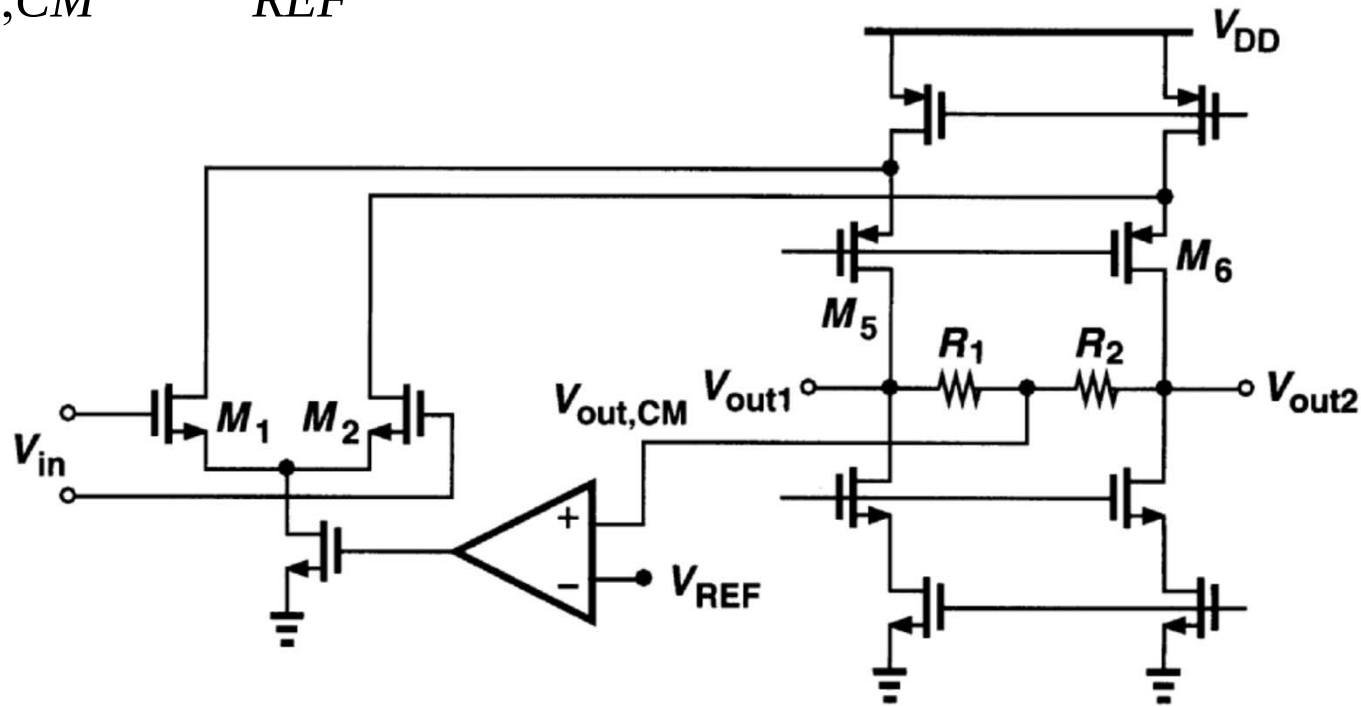
CMFB Example



Control the Bottom current source

Alternative CMFB for Folded Cascode

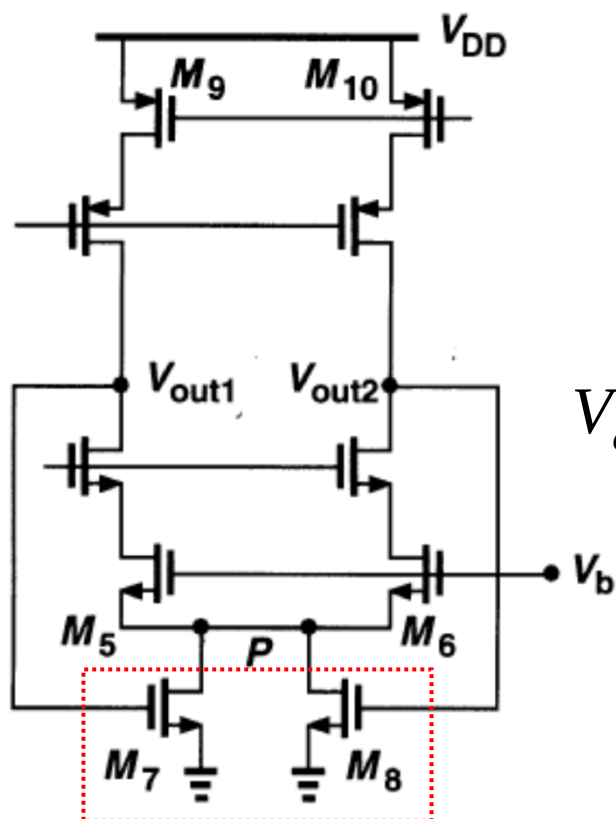
$$V_{out,CM} = V_{REF}$$



Alternative method of controlling output CM level



Simplified CMFB with Triode Devices



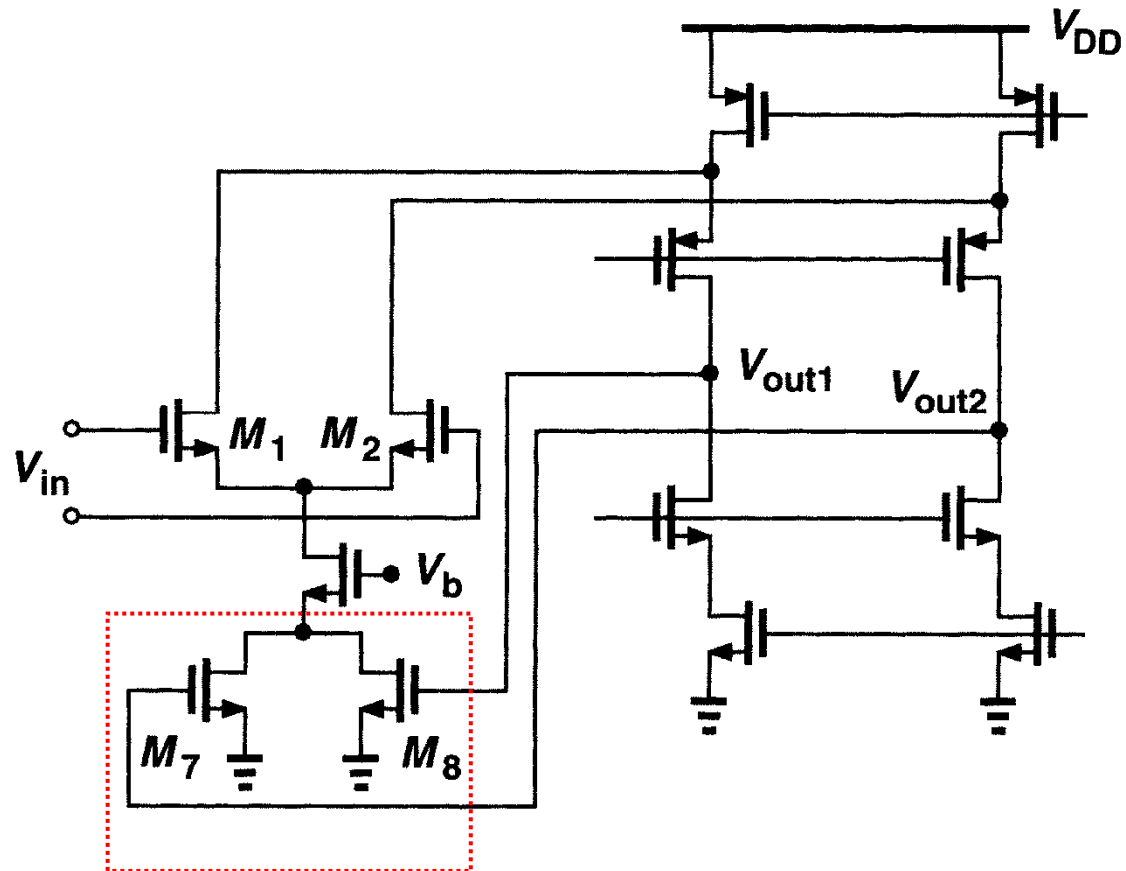
$V_{out,CM}$ sets a resistance (or a current) such that $I_5 + I_6 = I_9 + I_{10}$

$$V_{out1} + V_{out2} = \frac{2I_D}{\mu_n C_{OX} \left(\frac{W}{L}\right)_{7,8}} \frac{1}{V_b - V_{GS5}} + 2V_{TH}$$

Drawbacks:

1. CM level is a function of device parameters
2. Voltage drop on M7, M8 limit output swing
3. Increased output cap

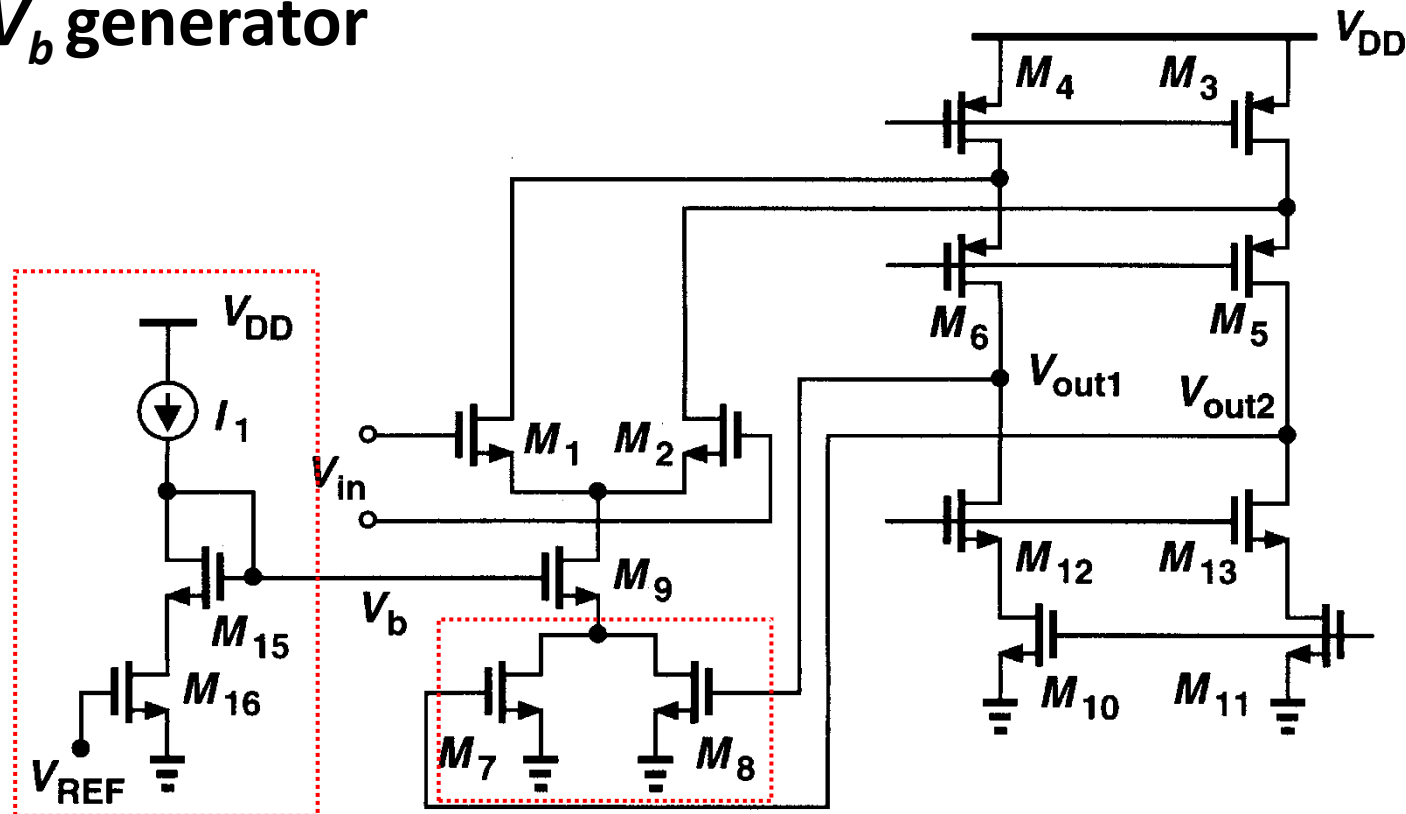
- Increase the output swing



Example: Triode CMFB with Reference

Modification for more accurate definition of output CM level

V_b generator

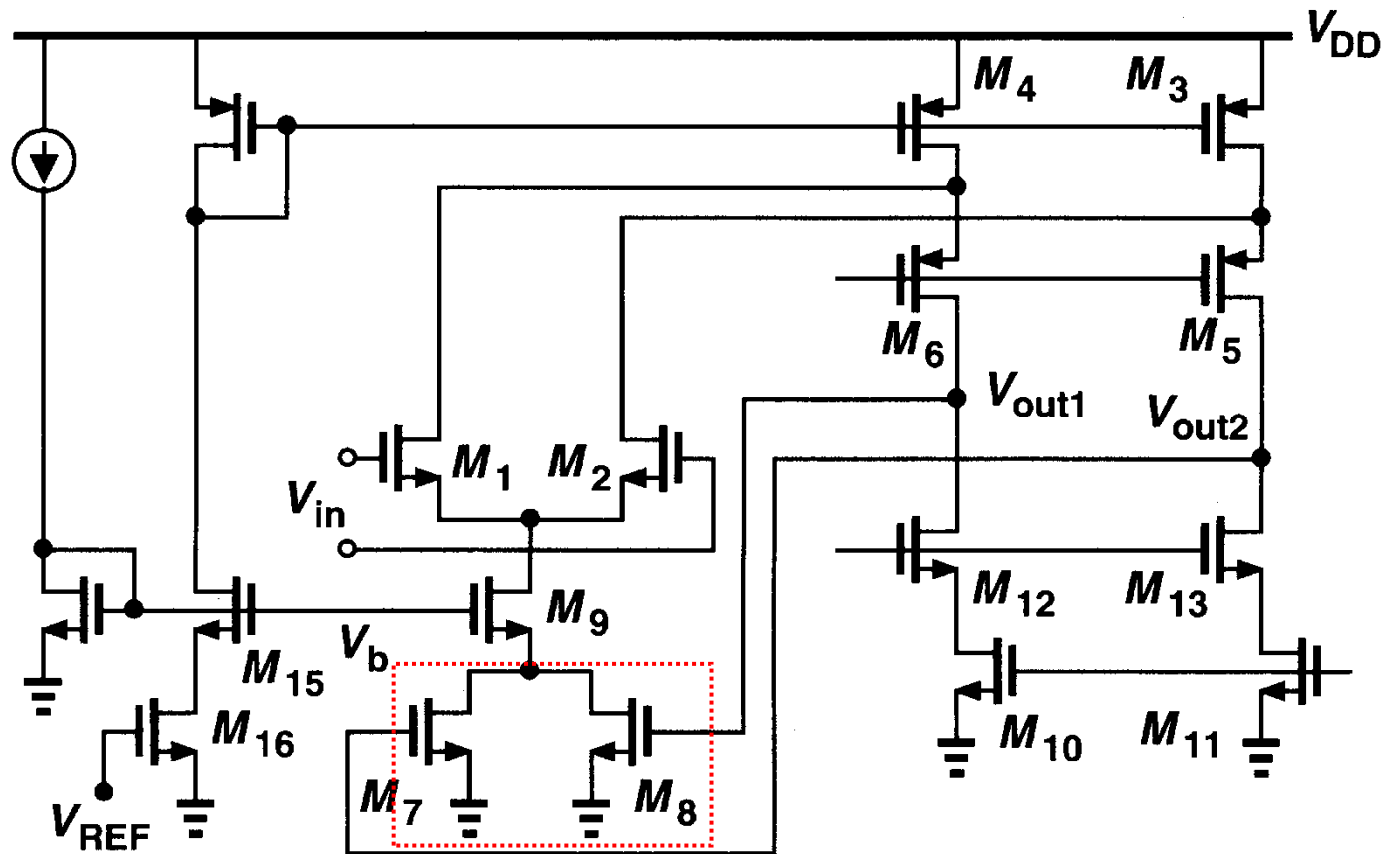


$$(W/L)_{15} = (W/L)_9$$

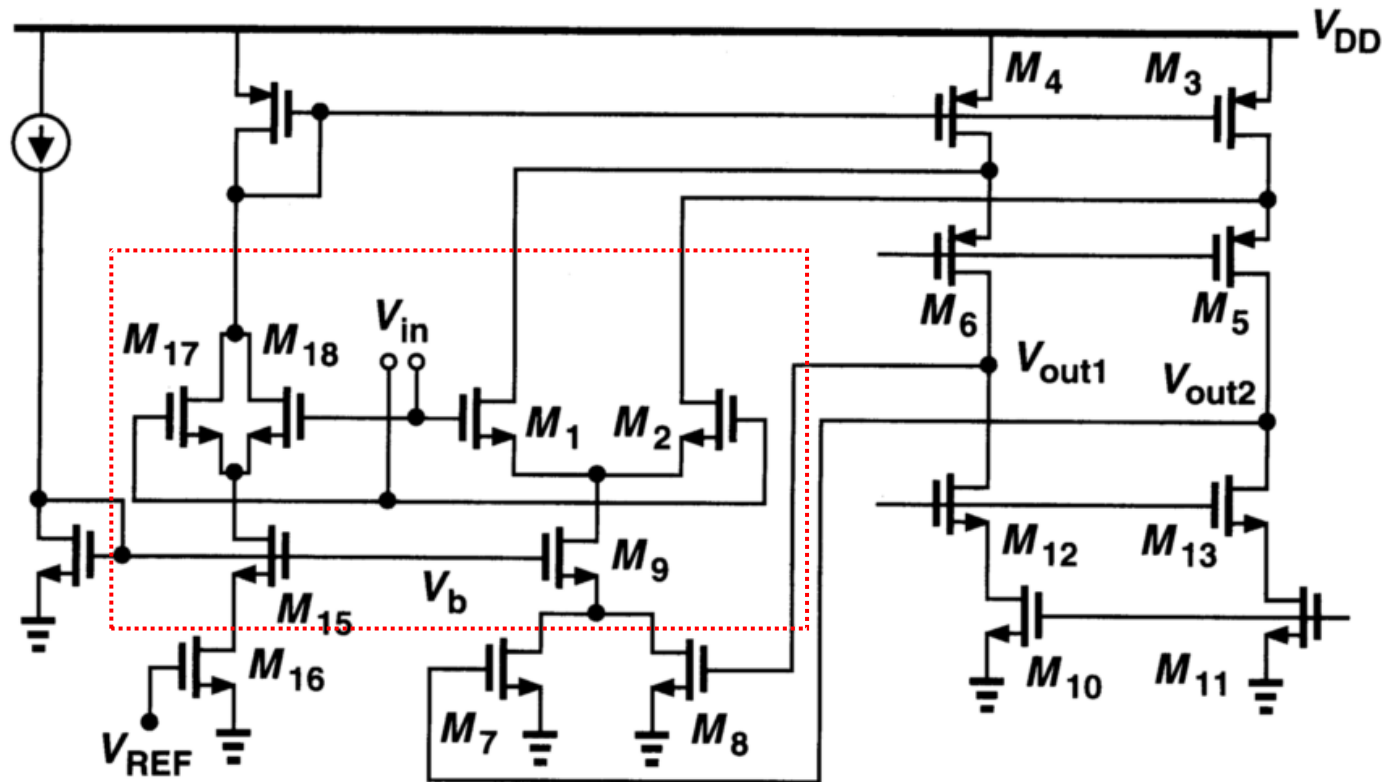
$$(W/L)_{16} = (W/L)_7 + (W/L)_8$$

$$V_{out,CM} = V_{REF}$$

- Completed structure



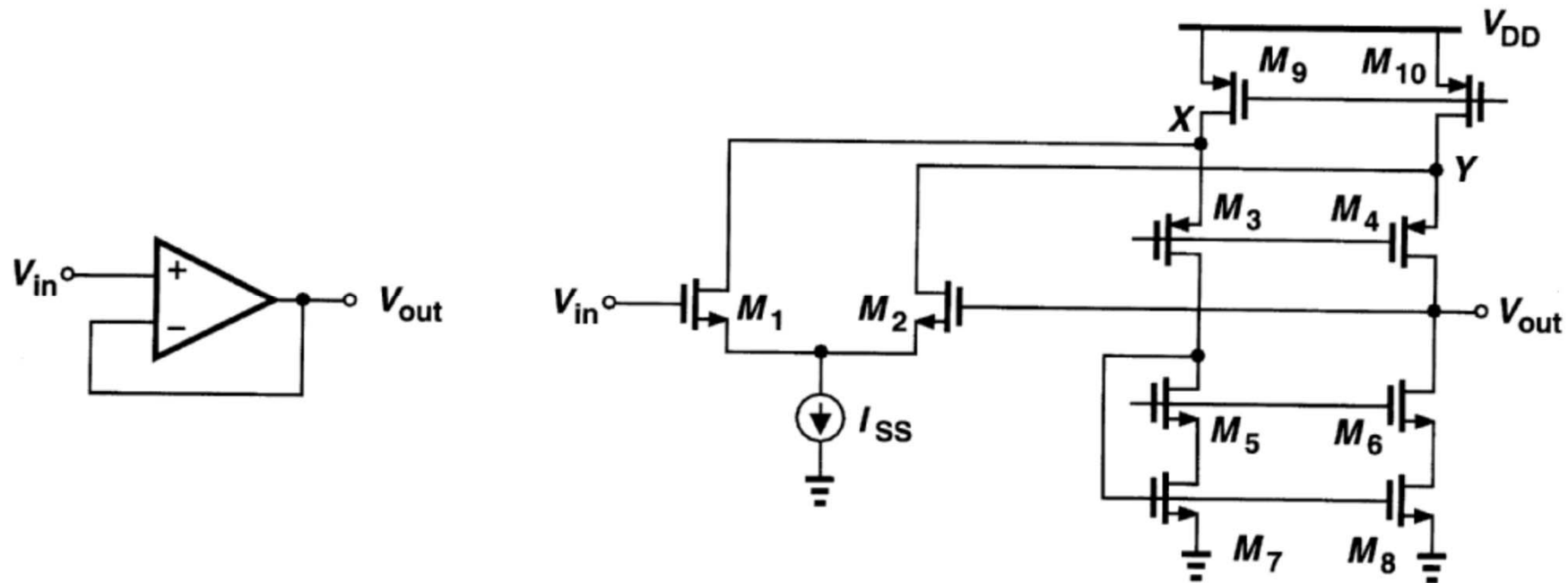
Modification to suppress error due to channel-length modulation.



Outline

- General consideration
- Single-stage OTA
- Two-Stage OTA
- Gain boosting technique
- Common-mode feedback
- **Input range limitations**
- Slew Rate
- Power supply rejection

Use folded cascode to obtain large input swing



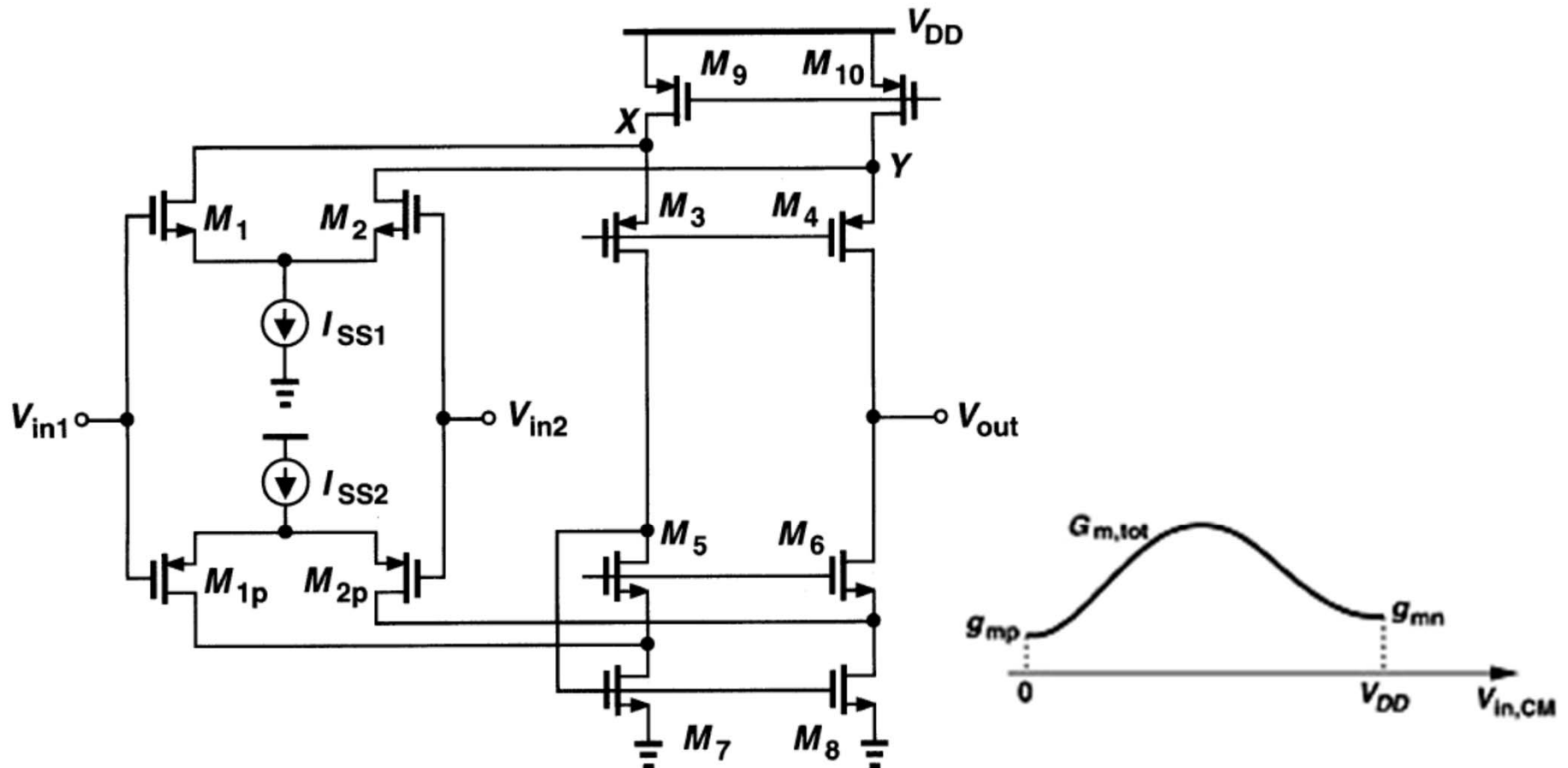
$$V_{in,min} = V_{OD,SS} + V_{GS1}$$

However, the minimum voltage at the output is $V_{OD6} + V_{OD8}$

How to extend the input range ?



Extension of input CM range



Outline

- General consideration
- Single-stage OTA
- Two-Stage OTA
- Gain boosting technique
- Common-mode feedback
- Input range limitations
- **Slew Rate**
- Power supply rejection

Response of a linear circuit to input step



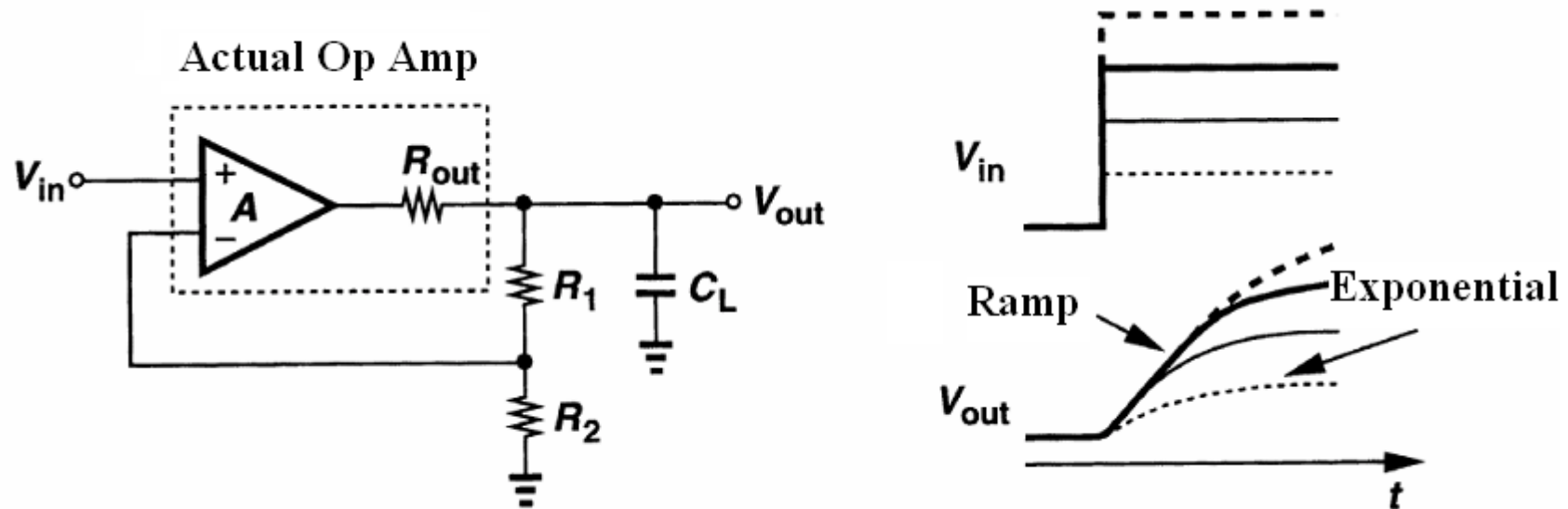
$$V_{out} = V_{in} \left[1 - \exp \frac{-t}{\tau} \right] \quad \tau = RC$$

$$\frac{dV_{out}}{dt} = \frac{V_{in}}{\tau} \exp \frac{-t}{\tau}$$

The Slop is a linear function of the step amplitude



Slewing in Op Amp



When V_{in} further increases, the slope keeps constant

→ We call the slope of the ramp the “**slew rate**”

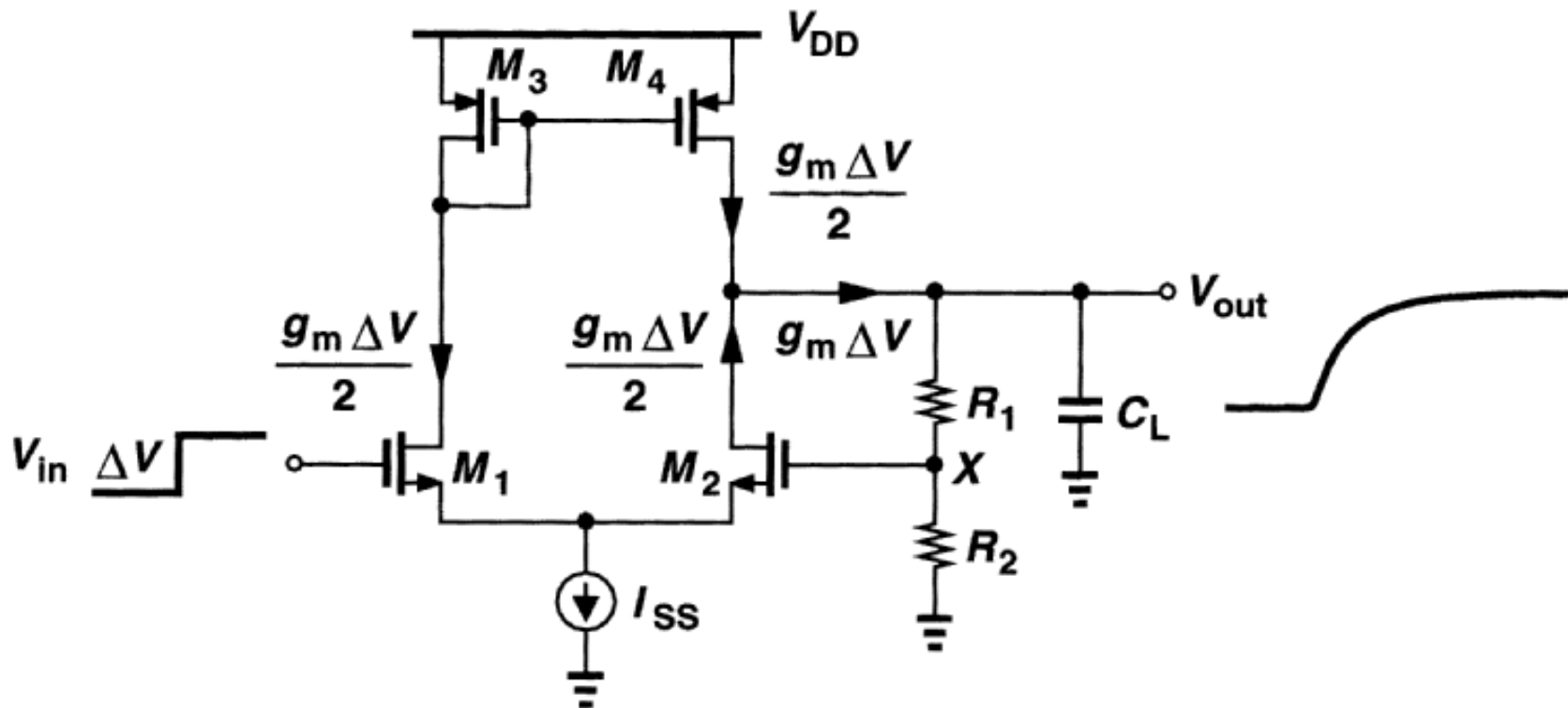
Why?



Small-Signal Operation of Op Amp

Small step:

The output current is $g_m \Delta V$, the slop is almost linear

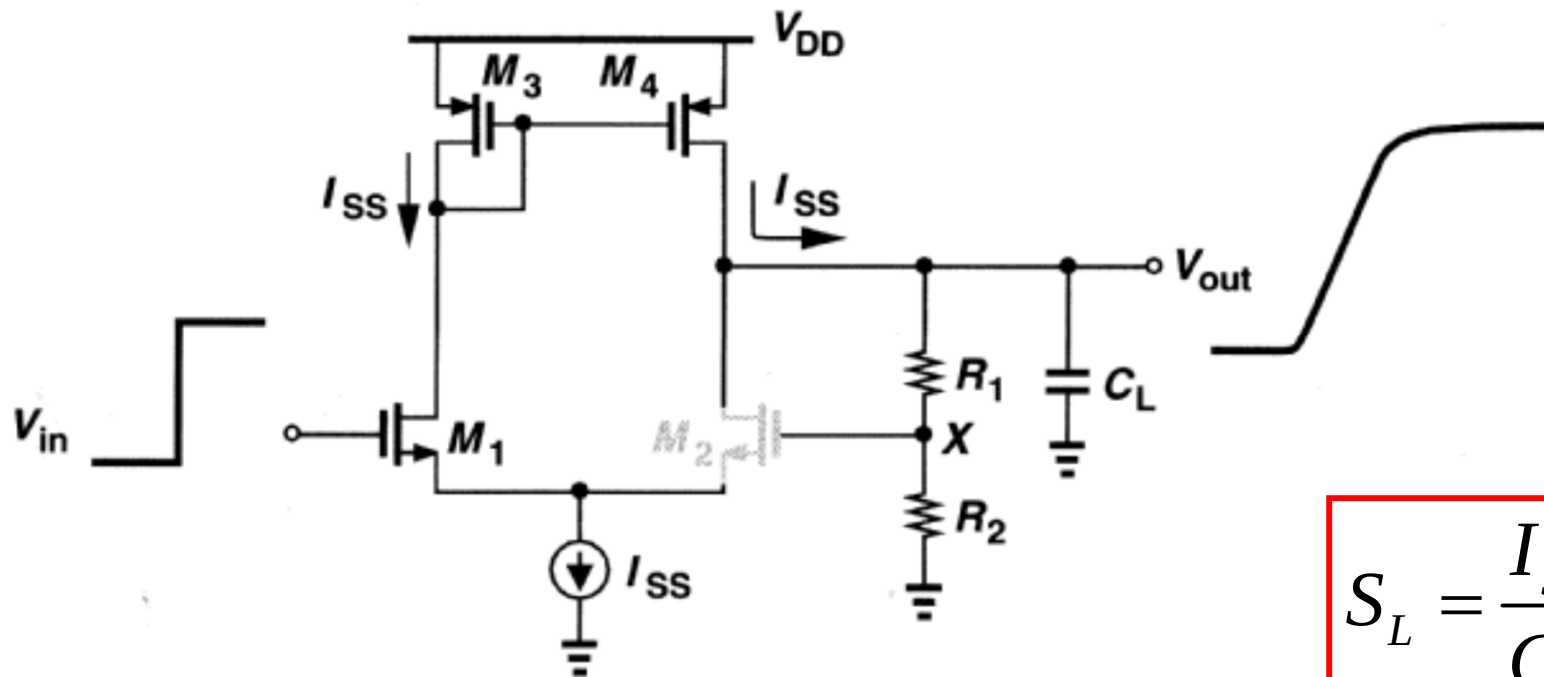


Op Amp Slewing: Large step

At beginning, the output current is a constant I_{SS}

→ Nonlinear

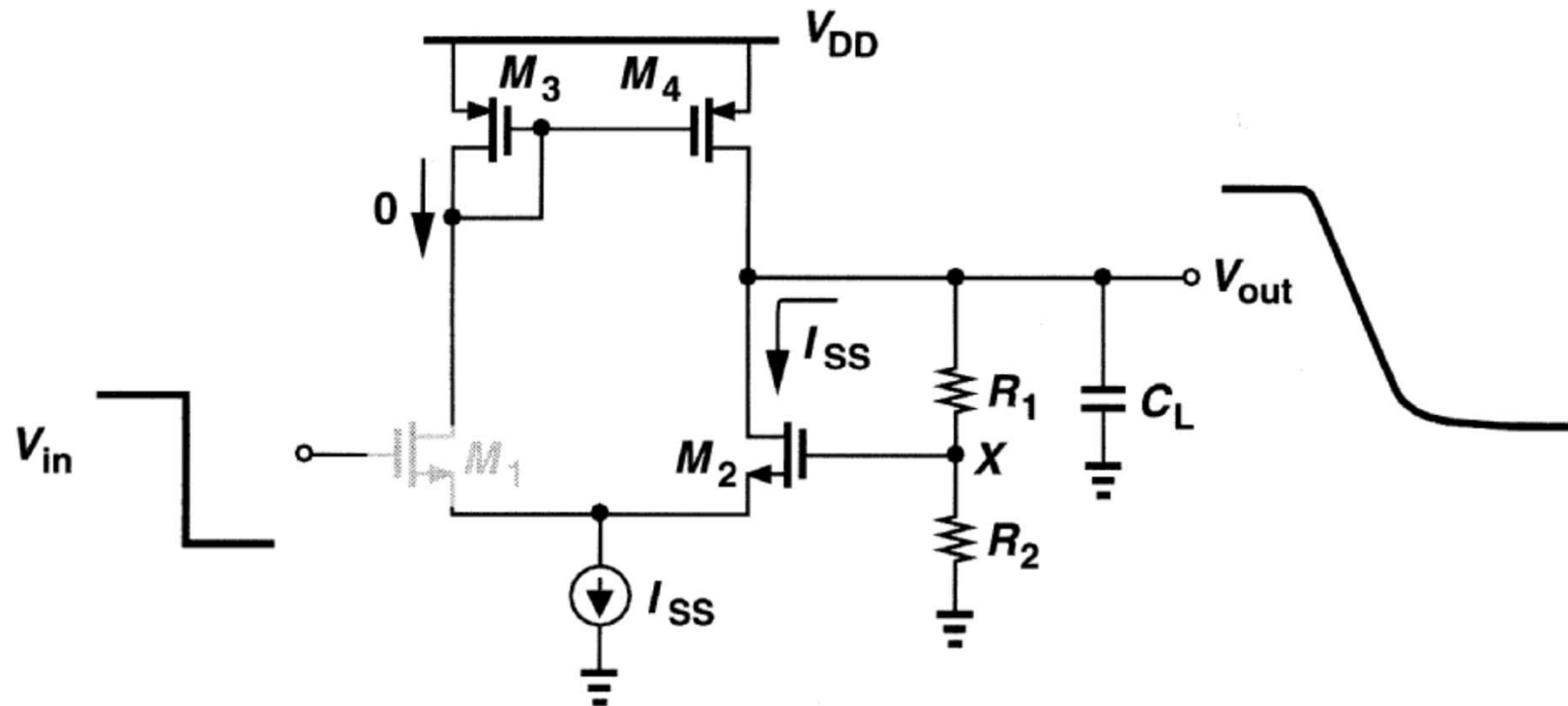
After M_2 is on, returns to linear behavior



$$S_L = \frac{I_{SS}}{C_L}$$



Slewing during high-to-low transition

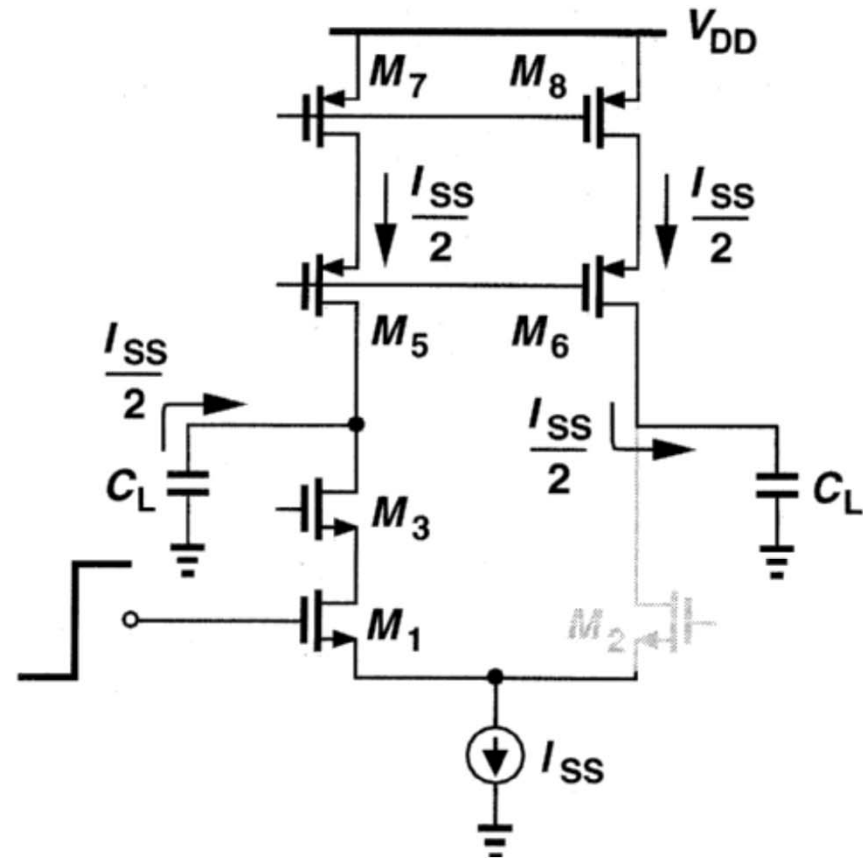
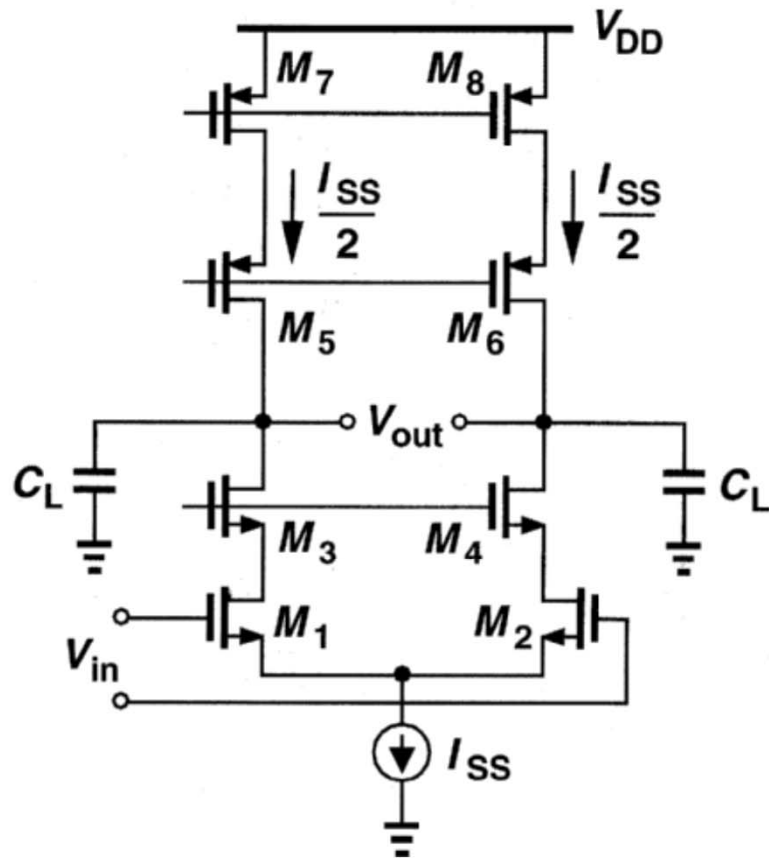


To amplify the signal $V_0 \sin \omega_0 t$ with no distortion

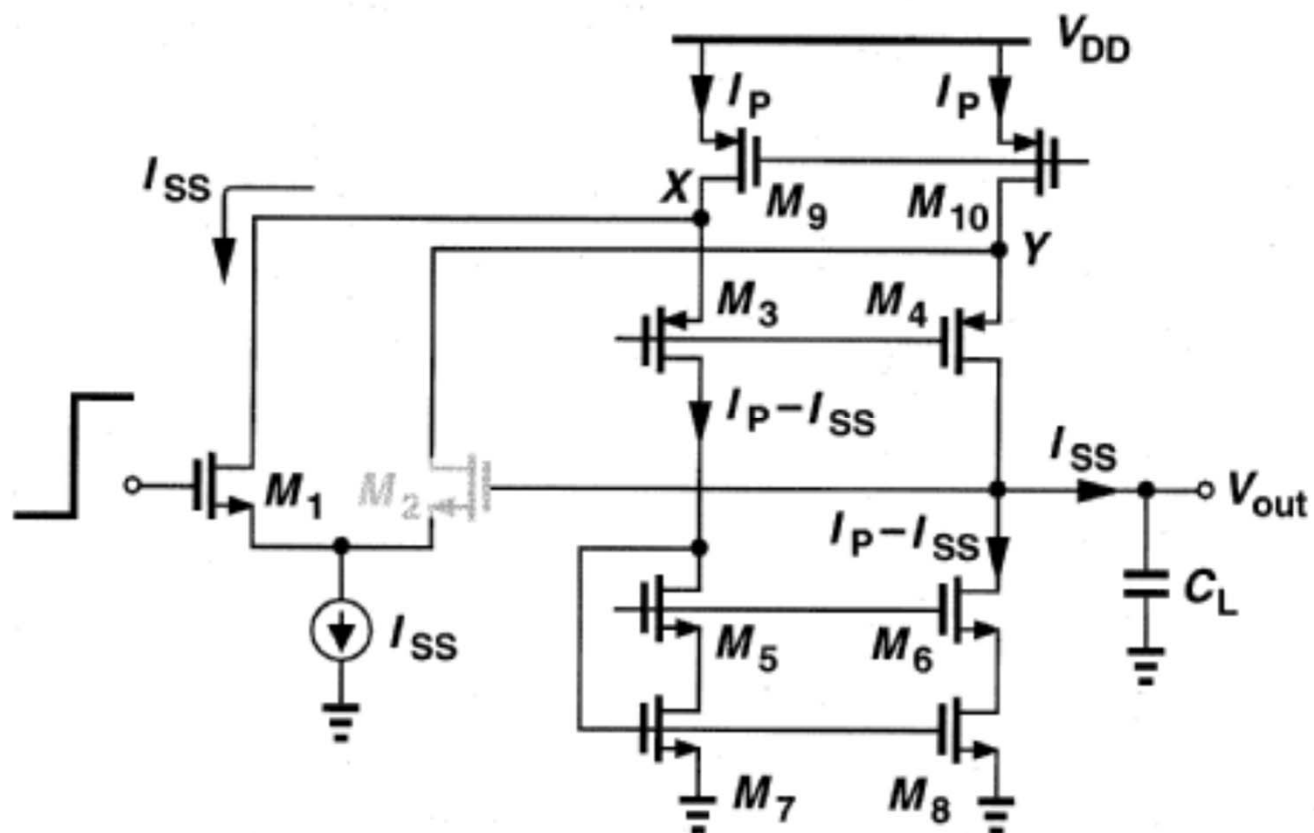
→ The required minimum slew rate is $\omega_0 V_0 (1 + R_1/R_2)$,

Why??

Slewing in fully-diff. telescopic Op Amp



Folded-Cascode Slewing



Folded-Cascode (cont.)

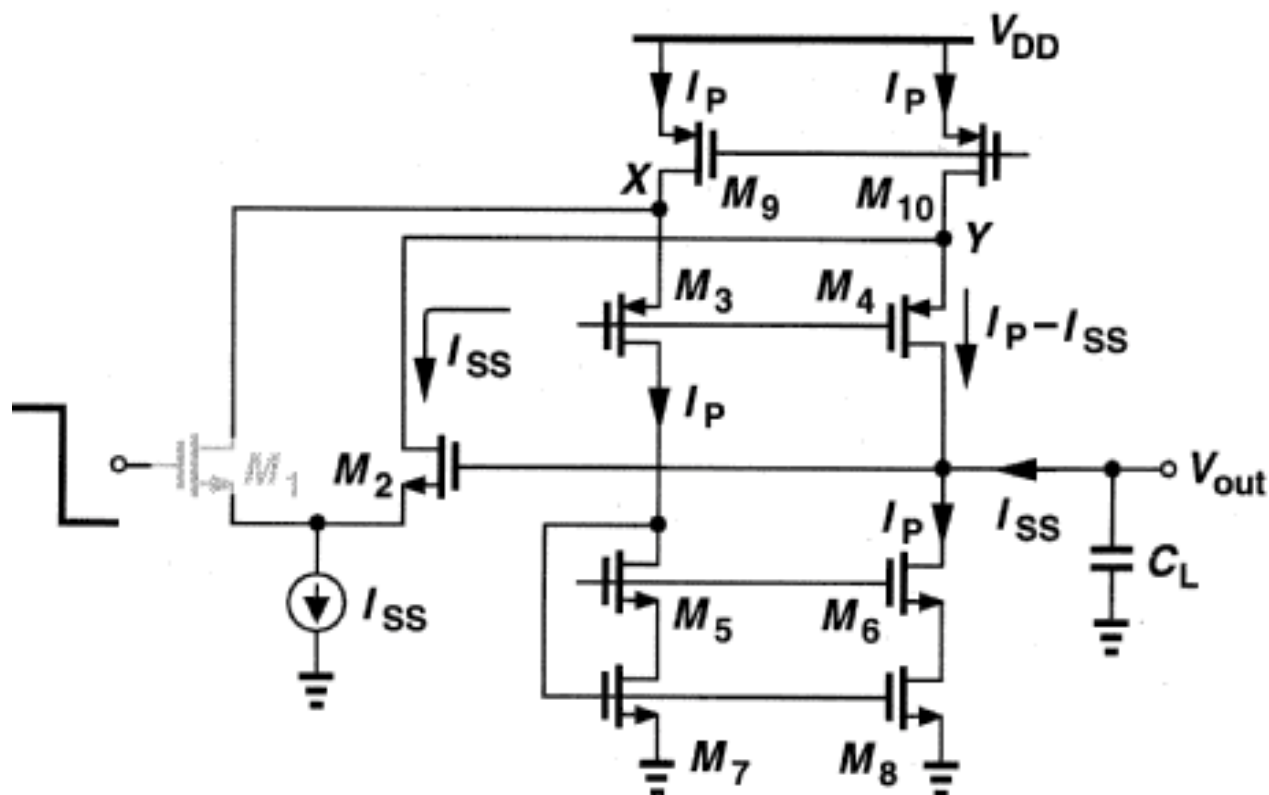


Fig. 9.58 Slewing in folding-cascode op amp

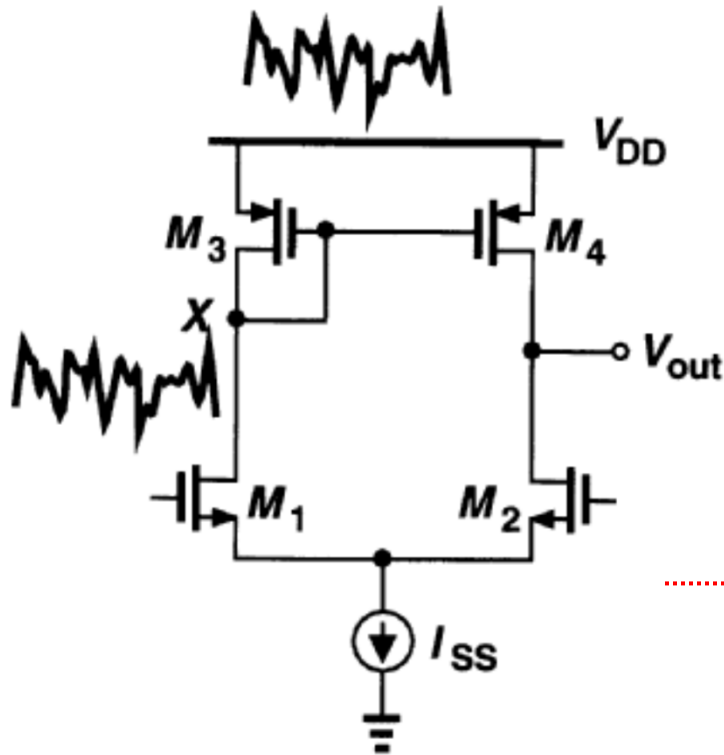
Outline

- General consideration
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- Two-Stage OTA
- Gain boosting technique
- Common-mode feedback
- Input range limitations
- Slew Rate
- Power supply rejection



Power Supply Rejection

Supply rejection of differential pair with active current mirror



$$PSRR = \frac{\frac{V_{out}}{V_{in}}(s)}{\frac{V_{out}}{V_{dd}}(s)}$$

$$PSRR \approx g_{mN}(r_{OP} \parallel r_{ON})$$



Summary

- General consideration
- Single-stage OTA
 - Differential pair
 - Telescope OTA
 - Folded cascode OTA
- Two-Stage OTA
- Gain boosting technique
- Common-mode feedback
- Input range limitations
- Slew Rate
- Power supply rejection

